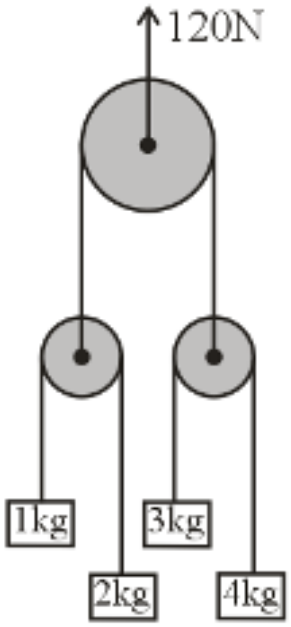
PHYSICS

1) In the given figure all pulleys are smooth and massless and strings are of negligible mass. Match



the information about block in List-I with results in List-II.

List-I		List-II	
(P)	acceleration of 1 kg block.	(1)	Zero
(Q)	acceleration of 2 kg block.	(2)	2.5 m/s^2
(R)	acceleration of 3 kg block.	(3)	5 m/s^2
(S)	acceleration of 4 kg block.	(4)	20 m/s^2

(1) P → 4; Q → 3; R → 1; S → 2

(2) P → 4; Q → 3; R → 2; S → 1

(3) P → 3; Q → 4; R → 1; S → 2

(4) P → 3; Q → 4; R → 2; S → 1

2)

The force exerted by the lift on the foot of a person standing in it, is more than his weight, then the lift is -
(a) going up and slowing down
(b) going up and speeding up
(c) going down and slowing down
(d) going down and speeding up

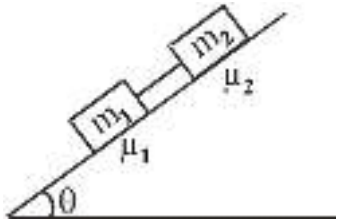
(1) a, b

(2) b, c

(3) c, d

(4) a,b,c,d

3) Two blocks are kept on an inclined plane and tied to each other with a mass-less string. Coefficient of friction between m_1 and inclined plane is μ_1 and that between m_2 and the inclined is μ_2,



Then :

(1) The tension in the string is zero, if $\mu_1 > \mu_2$

(2) The tension in the string is zero if $\mu_1 < \mu_2$

(3) Tension in the string is always zero irrespective of μ_1 and μ_2

(4) None of these

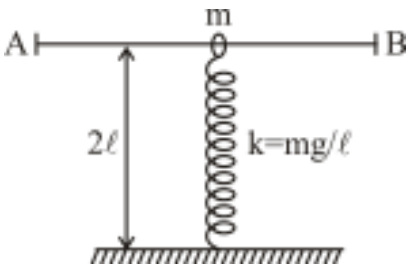
4) The momentum p (kg ms^{-1}) of a particle is varying with time (second) as $p = 4 + 5t^2$. The force on the particle at t = 2s is

(1) 12 N

(2) 16 N

(3) 18 N

(4) 20 N



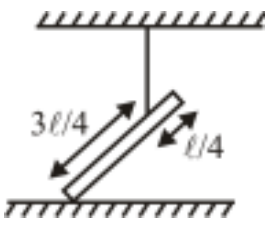
5)
A bead of mass m is free to move along the rod AB tied by a spring of natural length □/4. Find force on bead by rod.

(1) 2mg

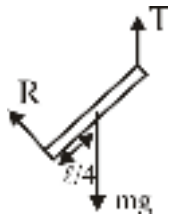
(2) mg

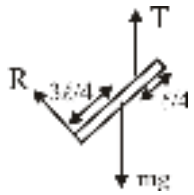
(3) $\frac{7mg}{4}$

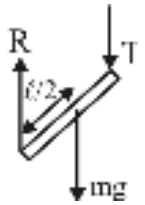
(4) $\frac{11mg}{4}$

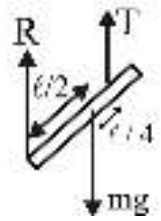


6) Which figure represent the correct FBD of rod of mass M

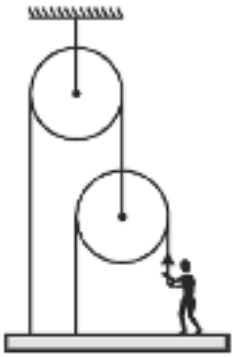
(1) 

(2) 

(3) 

(4) 

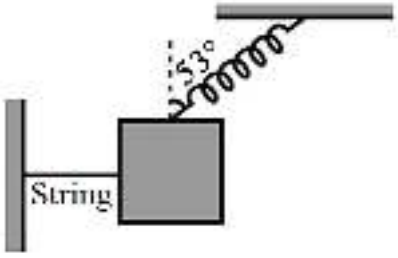
7) In the given diagram, with what force must the man pull the rope to hold the plank in position?



Weight of the man is 60 kg. Neglect the weights of plank, rope and pulley.

(1) 150 N
(2) 300 N
(3) 600 N
(4) 120 N

8) The block shown in the figure is in equilibrium. Find acceleration of the block just after the string is cut.



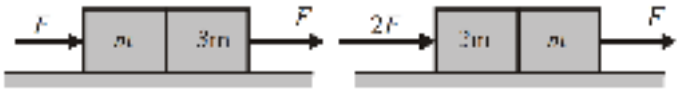
(1) 3g/5

(2) 4g/5

(3) 4g/3

(4) None

9) In both the given cases, when blocks in contact were at rest, the forces are applied as shown. All the surfaces are smooth. Which of the following statements is/are correct?



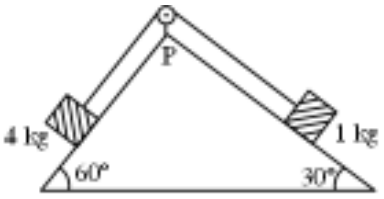
(1) Normal reaction between the blocks is zero in case I and 0.5F in case II.

(2) Normal reaction between the blocks is zero in case II and 0.5F in case I.

(3) Normal reaction between the blocks is zero in case I and F in case II.

(4) Normal reaction between the blocks is zero in case II and F in case I.

10) As per given figure, a weightless pulley P is attached on a double inclined frictionless surface.



The tension in the string (massless) will be (if g = 10 m/s^2)

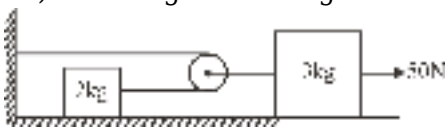
(1) $(4\sqrt{3}+1)$N

(2) $4(\sqrt{3}+1)$N

(3) $4(\sqrt{3}-1)$N

(4) $(4\sqrt{3}-1)$N

11) For the given arrangement if all surfaces are frictionless then acceleration of 2 kg block will be :



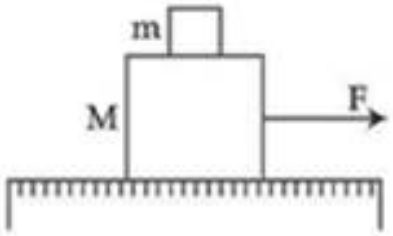
(1) $\frac{100}{11}$ m/s^2

(2) $\frac{100}{13}$ m/s^2

(3) $\frac{50}{11}$ m/s^2

(4) $\frac{50}{13}$ m/s^2

12) A system of two blocks of masses m = 2 kg and M = 8 kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be :



(1) 9.8 N

(2) 39.2 N

(3) 49 N

(4) 78.4 N

13) A marble block of mass 2 kg lying on ice when given a velocity of 6 m/s is stopped by friction in 10s. Then the coefficient of friction is:

(1) 0.02

(2) 0.03

(3) 0.06

(4) 0.01

14) A ball is suspended by a thread from the ceiling of a tram car. The brakes are applied and the speed of the car changes uniformly from 36 km h^{-1} to zero in 5s. The angle by which the ball deviates from the vertical is (g=10 m s^{-2})

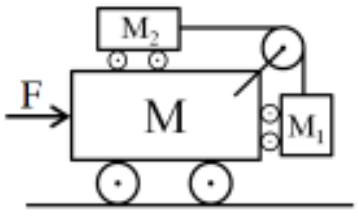
(1) $\tan^{-1}\left(\frac{1}{3}\right)$

(2) $\sin^{-1}\left(\frac{1}{5}\right)$

(3) $\tan^{-1}\left(\frac{1}{5}\right)$

(4) $\cot^{-1}\left(\frac{1}{3}\right)$

15) Find the force applied to the system if M_1 and M_2 remain at rest w.r.t. "M" :- (All surfaces are smooth)



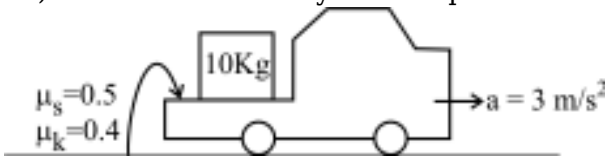
(1) $F=(M_1 + M_2 + M)$

(2) $F=(M_1 + M_2 + M)\frac{M_1}{M_2}g$

(3) $F = (M_1 + M_2 + M)\frac{M_2}{M_1}g$

(4) None of these

16) If block is stationary with respect to truck, then value of friction acting on the block is –



(1) 50 N

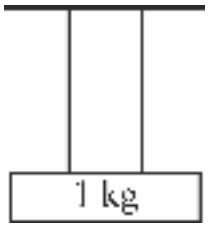
(2) 40 N

(3) 0

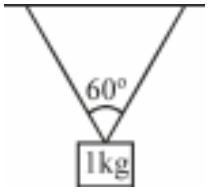
(4) 30 N

17) In which of the following figure tension in the string will be maximum :-

(1)



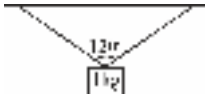
(2)



(3)



(4)



18) A spring of spring constant K is broken in the length ratio 1 : 3. The spring constant of larger part will be :

(1) $\frac{4K}{3}$

(2) $\frac{2K}{3}$

(3) $\frac{K}{3}$

(4) $\frac{5K}{3}$

19) An astronomical telescope has focal lengths 100cm & 10cm of objective and eyepiece lens respectively when final image is formed at least distance of distinct vision, magnification power of telescope will be :

(1) –10

(2) –11

(3) –14

(4) –15

20) The focal length of objective and eye lens of a microscope are 4 cm and 8 cm respectively. If the least distance of distinct vision is 24 cm and object distance is 4.5 cm from the objective lens, then the magnifying power of the microscope will be : (Final image is at infinity)

(1) 18

(2) 32

(3) 24

(4) 20

21) A microscope has an objective of focal length 1.5 cm and an eye-piece of focal length 2.5 cm. If the distance between objective and eye-piece is 25 cm the approximate value of magnification produced for relaxed eye is ?

(1) 75

(2) 110

(3) 140

(4) 25

22) Focal length of a magnifying glass is 12.5 cm. Ratio of maximum and minimum magnifying power is :-

(1) 2 : 3

(2) 1 : 3

(3) 3 : 1

(4) 3 : 2

23) An achromatic doublet of focal length 90 cm is to be made of two lens. The material of one having 1.5 times the dispersive power of other The doublet is converging type find focal length of each lens:-

(1) 25 cm, –15 cm

(2) 30 cm, – 45 cm

(3) 10 cm, –15 cm

(4) 5 cm, –50 cm

24) An object and a screen are placed at a distance of 60 cm. A convex lens is to be placed between them. For what minimum value of focal length of convex lens, we may get sharp image on screen ?

(1) 15 cm

(2) 20 cm

(3) 25 cm

(4) 30 cm

25) An object is put at a distance of 5 cm from the first focus of a convex lens of focal length 10cm.

If a real image is formed it's distance from the lens will be :-

(1) 15 cm
(2) 20 cm
(3) 25 cm
(4) 30 cm

26) The focal length of a convex lens of glass (μ = 1.5) is 2 cm. The focal length of the lens when immered in a liquid of refractive index 1.25 will be :-

(1) 5 cm
(2) 2.4 cm
(3) 1 cm
(4) 4 cm

27) A doctor advises a patient to use spectacles with a convex lens of focal length 40 cm in contact with a concave lens of focal length 25 cm. What is the power of the resultant combination–

(1) 1.5 D
(2) –1.5 D
(3) 6.5 D
(4) –6.5 D

28) The distance between a real object and a divergent lens is m times the focal length of the lens. The linear magnification produced by the lens is :-

(1) m
(2) $\frac{1}{m}$
(3) m + 1
(4) $\frac{1}{m+1}$

29) A parallel beam of light travelling in water (refractive index = 4/3) is refracted by a spherical air bubble of radius 2cm situated in water. Assuming the light rays to be paraxial, the position of the image due to refraction at the first surface is -

(1) 6 cm from the first surface
(2) 12 cm from the first surface
(3) 3 cm from the first surface
(4) 10 cm from the first surface

30) The angle of a prism is ‘A’. One of its refracting surfaces is silvered. Light rays falling at an angle of incidence 2A on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ, of the prism is :-

(1) 2 sin A

(2) 2 cos A

(3) $\frac{1}{2}$ cosA

(4) tanA

31) A ray of light passes through equilateral Prism (μ =1.5) such that angle of incidence is equal to angle of emergence and emergence angle is equal to 3/4th of Prism angle. The angle of deviation is :-

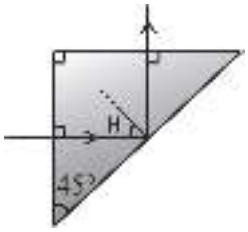
(1) 60°

(2) 30°

(3) 45°

(4) 120°

32) A triangular prism of glass is shown in the figure. A ray incident normally to one face is totally reflected, if θ = 45°. The index of refraction of glass is :-



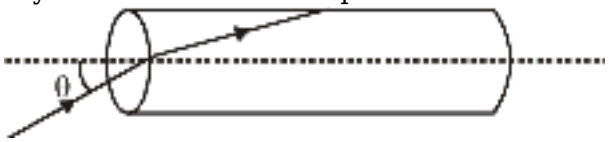
(1) Less than 1.41

(2) Equal to 1.41

(3) Greater than 1.41

(4) None of the above

33) A transparent solid cylindrical rod has a refractive index of $2/\sqrt{3}$. It is surrounded by air. A light ray is incident at the midpoint of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is :

(1) $\sin^{-1}(1/2)$

(2) $\sin^{-1}(\sqrt{2}/2)$

(3) $\sin^{-1}(2/\sqrt{3})$

(4) $\sin^{-1}(1/\sqrt{3})$

34) The velocity of light in a transparent liquid medium is 1.8×10^8 m/sec, while that in vaccum is 3×10^8 m/sec. Find by how much the bottom of the vessel containing the liquid appears to be raised if the depth of the liquid is 0.25m :-

(1) 0.1 m

(2) 0.15 m

(3) 0.416 m

(4) 0.20 m

35) A point object is placed at a distance of 30 cm from a convex mirror of focal length 30 cm. The image will form at :-

(1) Infinity

(2) Pole

(3) Focus

(4) 15 cm behind the mirror

36) For real object match the corresponding entries of column-1 with coloumn-2 (where m is the magnification produced by the mirror) :-

	Column-1		**Column-2**
A	m = –4	a	Convex mirror
B	m = –1/4	b	Concave mirror
C	m = +3	c	Real image
D	m = +1/3	d	Virtual image

(1) A-a and c, B-a and d, C-a and b, D-c and d

(2) A-a and d, B-b and c, C-b and d, D-b and d

(3) A-c and d, B-b and d, C-b and c, D-a and d

(4) A-b and c, B-b and c, C-b and d, D-a and d

37) θ_x and θ_y are the angles made by a vector $\vec{A}$ with positive x and positive y-axis respectively. Which set of θ_x and θ_y is /are not possible?
(I) 60°, 60° (II) 45°, 60°
(III) 30°, 45° (IV) 30°, 30°

(1) Only III

(2) Only IV

(3) Only II and III

(4) Only III and IV

38) **Assertion (A) :-** All coinitial vectors are coplanar.
Reason (R) :- Coplanar vectors must lie in the same plane.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct and R is not the correct explanation of A

(3) A is correct and R is wrong

(4) A is wrong and R is correct

39) If $\vec{A} = 2\hat{i} + \sqrt{7}\hat{j}$ and $\vec{B} = 5\hat{i} + \sqrt{7}\hat{j} - 3\hat{k}$ then the vector whose magnitude is equal to $\vec{A} \cdot \vec{B}$ and parallel to $\vec{B} - \vec{A}$ is :

(1) $\frac{17}{\sqrt{2}}\left(\hat{k}-\hat{j}\right)$

(2) $\frac{17}{\sqrt{2}}\left(\hat{i}-\hat{k}\right)$

(3) $3\hat{i}-3\hat{k}$

(4) $3\hat{k}-3\hat{i}$

40) $\vec{A}, \vec{B}$ and $\vec{C}$ are vectors each having a unit magnitude. If $\vec{A}+\vec{B}+\vec{C}=\vec{0}$, $\vec{A}.\vec{B}+\vec{B}.\vec{C}+\vec{C}.\vec{A}$ will be:-

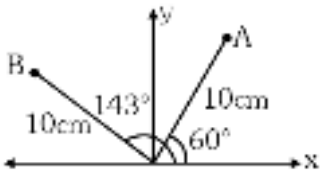
(1) 1

(2) $-\frac{3}{2}$

(3) $-\frac{1}{2}$

(4) 0

41) Refer the given figure and identify incorrect statement



(1) Distance of A from x-axis is 5√3 cm.

(2) Distance of B from x-axis is 6 cm.

(3) Distance of A from y-axis is 5 cm.

(4) Distance of B from y-axis is 6 cm.

42) Two non zero vectors $\vec{a}$ and $\vec{b}$ are such that $\left|\vec{a}+\vec{b}\right|=\left|\vec{a}-2\vec{b}\right|$ then $\frac{|\vec{a}|}{|\vec{b}|}$ may be :

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{1}{3}$

(4) $\frac{2}{3}$

43) The vector sum of three forces having magnitude $|\vec{F}_1|$ = 10 N, $|\vec{F}_2|$ = 8 N and $|\vec{F}_3|$ = 6 N acting on a particle is zero. Then angle between $\vec{F}_1$ and $\vec{F}_2$ will be :-

(1) 53°

(2) 127°

(3) 37°

(4) 143°

44) If a unit vector is represented by $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$ then the value of 'c' is:-

(1) 1

(2) $\frac{1}{\sqrt{5}}$

(3) $\sqrt{0.11}$

(4) $\sqrt{0.01}$

45) The value of the constant P such that the vectors $(2\hat{i} - \hat{j} + \hat{k})$, $(\hat{i} + 2\hat{j} - 3\hat{k})$ & $(3\hat{i} + p\hat{j} + 5\hat{k})$ are coplanar is :-

(1) 0

(2) 2

(3) –4

(4) –2

CHEMISTRY

1) Which one of the following statements regarding Henry's law is not correct ?

(1) The value of K_H increases with increase of temperatrue and K_H is function of the nature of the gas

(2) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids.

(3) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.

(4) Different gases have different K_H (Henry's law constant) values at the same temperature.

2) If Ar gas is bubbled through water at 298 K, how many moles of Ar gas would dissolve in 1 litre of water? Assume that Ar exerts a partial pressure of 0.403 bar. Given that Henry's law constant for Ar at 298 K is 40.3 K bar

(1) 6.02×10^{-3}

(2) 1×10^{-5}

(3) 18×10^{-5}

(4) 55.56×10^{-5}

3) Two liquids A and B have, P_A^0 and P_B^0 = 1 : 3 at a certain temperature. Assume A and B form an ideal solution, and the ratio of mole of A to B in the liquid phase are 1 : 3, then mole fraction of A in vapour phase in equilibrium with the solution -

(1) 0.1

(2) 0.2

(3) 0.5

(4) 1.0

4) The vapour pressure of a solution of a non-volatile electrolyte B in a solvent A is 95% of the

vapour pressure of the solvent at the same temperature. If the molecular weight of the solvent is 0.3 times the molecular weight of solute, the weight ratio of the solvent and solute are

(1) 0.15

(2) 5.7

(3) 0.2

(4) 4.0

5) If P_0 and P_S are the vapour pressure of solvent and its solution respectively. N_1 and N_2 are the mole fraction of solvent and solute respectively then

(1) $P_S = \frac{P_0}{N_2}$

(2) $P_0 - P_S = P_0N_2$

(3) $P_S = P_0N_2$

(4) $\frac{(P_0 - P_S)}{P_S} = \frac{N_1}{(N_1 + N_2)}$

6) Find weight in gram of glucose (M_w = 180) that should be dissolved in 100 gm water in order to produce a solution with a difference of 105ºC between its boiling point and freezing point temperatures.
(Take K_f = 2 K-kg/mole, K_b = 0.5 K-kg/mole)

(1) 360 gm

(2) 72 gm

(3) 36 gm

(4) 180 gm

7) Given below are two statements :
Statement (I) : NaCl is added to the ice at 0°C, present in the ice cream box to prevent the melting of ice cream.
Statement (II) : On addition of NaCl to ice at 0°C, there is a depression in freezing point. In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement I** is false but **Statement II** is true

(2) Both **Statement I** and **Statement II** are true

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is true but **Statement II** is false

8) **Assertion (A) :** When methyl alcohol is added to water, boiling point of water increases.
Reason (R) : When a non-volatile solute is added to a volatile solvent, elevation in boiling point is observed

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

9) The colligative property used for the determination of molar mass of polymers and proteins is

(1) depression in freezing point

(2) relative lowering in vapour pressure

(3) elevation in boiling point

(4) osmotic pressure

10) Boiling point of dilute liquid solution containing non-volatile solute in different solvent depends on :
(I) Concentration of non volatile solute in the solution
(II) Boiling point of pure solvent
(III) Enthalpy of vapourisation of pure solvent
(IV) Molar mass of solvent

(1) I, II, III

(2) II, III, IV

(3) I, IV

(4) I, II, III, IV

11) 0.1 moles of a non-volatile, non-electrolyte solute in dilute solution dissolved in same mass of $CCl_4 (T_b^0 = 350\,K;\ K_b = 5\,K\,Kg\,mol^{-1})$ & $CS_2\ (T_b^0 = 319\,K;\ K_b = 2.3\,K\,Kg\,mol^{-1})$ solvent to form two different dilute solutions. Identify the correct statement.

(1) Both the solutions will have same osmotic pressure at same temperature.

(2) Both the solutions will have same boiling point.

(3) Both the solutions will have same elevation of boiling point.

(4) Solution with CCl_4 as solvent will have higher boiling point.

12) A dilute solution having one mole of solute 'S' in 1 Kg solvent with molal elevation constant k_b is taken. Solute dimerises in the solvent. The degree of association 'α' is given as. (ΔT = Elevation in B.P.)

(1) $\alpha = \dfrac{\Delta T}{2k_b}$

(2) $\alpha = \dfrac{2 + 2\Delta T}{k_b}$

(3) $\alpha = \dfrac{k_b - \Delta T}{2\Delta T}$

(4) $\alpha = \dfrac{2(k_b - \Delta T)}{k_b}$

13) Match the following list for the given solutions prepared by mixing different components :

List-I		List-II	
(P)	Acetone + $CHCl_3$	(1)	$\Delta S_{mix.} > 0$

(Q)	Ethanol + Water	(2)	$\Delta V_{mix.} > 0$
(R)	$C_2H_5Br + C_2H_5I$	(3)	$\Delta H_{mix.} < 0$
(S)	Acetone + Benzene	(4)	Minimum boiling azeotrope

(1) P-2, Q-3, R-4, S-1

(2) P-4, Q-2, R-1, S-3

(3) P-2, Q-3, R-1, S-4

(4) P-3, Q-2, R-1, S-4

14) The freezing point of a diluted milk sample is found to be –0.2°C, while it should have been –0.5°C for pure milk. How much water has been added to pure milk to make the diluted sample ?

(1) 2 cups of water to 3 cups of pure milk

(2) 1 cup of water to 3 cups of pure milk

(3) 3 cups of water to 2 cups of pure milk

(4) 1 cup of water to 2 cups of pure milk

15) At 35°C, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is :

(1) heat must be absorbed in order to produce the solution at 35°C

(2) Raoult's law is not obeyed by this system

(3) a mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL

(4) CS_2 and acetone are less attracted to each other than to themselves

16) Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous solution of a non-volatile solute B. The relation between molecular weights of A and B is.

(1) $M_A = 4M_B$

(2) $M_B = 4M_A$

(3) $M_A = 8M_B$

(4) $M_B = 8M_A$

17) At a given temperature, the vapour pressure of a solution of two volatile liquids A and B is given by the equation
$P_S = 120 - 80X_B$ (X_B = mole fraction of B)
Vapour pressure of pure A and B at the same temperature are respectively :-

(1) 120, 80

(2) 120, 200

(3) 120, 40

(4) 80, 40

18) If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :

(1) 1 molar

(2) 0.1 molar

(3) 0.5 molar

(4) 0.1 molal

19) Mole fraction of glycerine($C_3H_5(OH)_3$) in a solution of 36 g of water and 46 g of glycerine is

(1) 0.46

(2) 0.36

(3) 0.20

(4) 0.40

20) H_2O_2 solution used for hair bleaching is sold as a solution of approximately 5.0 g H_2O_2 per 100 mL of the solution. The molecular mass of H_2O_2 is 34. The molarity of this solution is approximately:-

(1) 0.15 M

(2) 1.5 M

(3) 3.0 M

(4) 3.4 M

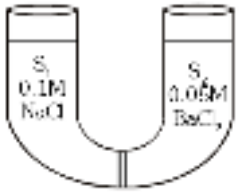
21) Which of the following has been arranged in order of decreasing freezing point :-

(1) 0.05 M KNO_3 > 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$

(2) 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$ > 0.05 M KNO_3

(3) 0.075 M $CuSO_4$ > 0.140 M sucrose > 0.04 M $BaCl_2$ > 0.05 M KNO_3

(4) 0.075 M $CuSO_4$ > 0.05 M KNO_3 > 0.140 M sucrose > 0.04 M $BaCl_2$

22) Two solutions S_1 and S_2 containing 0.1 M NaCl (aq) and 0.05 M $BaCl_2$ (aq) are separated by semiperameable membrane. 

Which among the following statement(s) is/are correct ? (Assume complete dissociation of both the electrolytes)

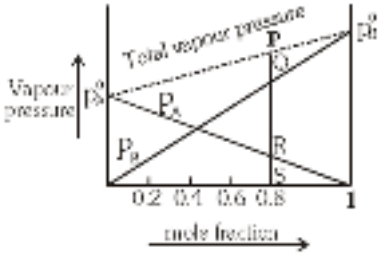
(1) S_1 and S_2 are isotonic

(2) S_1 is hypertonic while S_2 is hypotonic

(3) S_1 is hypotonic while S_2 is hypertonic

(4) All the above

23) Consider the following vapour pressure-composition graph, SP is equal to :-



(1) PQ + RS

(2) QR + RS

(3) SR + SQ

(4) PQ + QR

24) A solution containing 0.1 g of a non-voltatile organic substance P(molecular mass 100) in 100g of benzene raises the boiling point of benzene by 0.2°C, while a solution containing 0.1 g of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by 0.4°C. What is the ratio of molecular masses at P and Q ?

(1) 1 : 2

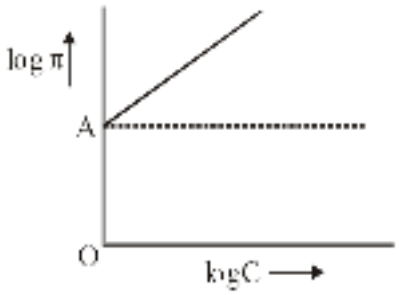
(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

25)

Variation of logπ (atm) v/s logC
(C = concentration of solute in g/L) is given



At 300K, for a solute of molar mass 300 g/mol, OA = log ($x \times 10^{-2}$).
What is the value of x ?

(1) 8.21

(2) 0.0821

(3) 82.1

(4) 0.821

26) A 0.025 M solution of monobasic acid had a freezing point of –0.06°C. Calculate K_a for the acid. K_f for H_2O is 1.86°C molality^{-1}. Assume molality equal to molarity.

(1) 1.96×10^{-3}

(2) 3.96×10^{-3}

(3) 2.96×10^{-3}

(4) 5.8×10^{-3}

27)

How much water is needed to dilute 10 ml of 10 N hydrochloric acid to make it exactly decinormal (0.1 N) :-

(1) 990 ml
(2) 1000 ml
(3) 1010 ml
(4) 100 ml

28) The vapour pressure of a solvent A is 0.80 atm. When a non-volatile solute B is added to this solvent its vapour pressure drops to 0.6 atm. What is the mole fraction of B in solution :-

(1) 0.25
(2) 0.50
(3) 0.75
(4) 0.90

29) Out of molarity (M), molality (m), formality (F) and mole fraction (x), those independent of temperature are :-

(1) M, m
(2) F, x
(3) M, x
(4) m, x

30)

Which of the following shows positive deviation from Raoult's law :
(i) Chloroform and acetone
(ii) Carbondisulphide and acetone
(iii) Ethanol and Acetone
(iv) Phenol and Aniline

(1) i and ii
(2) ii and iii
(3) i and iv
(4) ii and iv

31) A solution of glucose is isotonic with 4g of urea per litre of solution. The concentration of glucose is :-

(1) 4 g/L
(2) 8 g/L
(3) 12 g/L

(4) 14 g/L

32) Calculate the percentage dissociation of an electrolyte XY_2 (Normal molar mass = 250) in water if the observed molar mass by measuring elevation in boiling point is 100 :-

(1) 75%

(2) 25%

(3) 65%

(4) None of these

33) Which of the following is not correct for non-ideal solutions ?

(1) $P_S < x_A P_A^0 + x_B P_B^0$

(2) ΔH_{mixing} = –ve

(3) $P_S = P_B^0 x_B + P_A^0 x_A$

(4) $\Delta V_{mixing} > 0$

34) 1.1g of $CoCl_3.6NH_3$ (M_w = 267) was dissolved in 100g of H_2O. The freezing point of the solution was – 0.29°C. How many moles of solute particles exist in solution for each mole of solute introduced nearly (k_f for water = 1.86) :-

(1) 1.75

(2) 2

(3) 3

(4) 4

35) On mixing 30 mL CCl_4 and 10 mL C_6H_6, the total volume of the solution is :-

(1) > 40 mL

(2) < 40 mL

(3) = 40 mL

(4) Can't be predicted

36) Which of the following is correct regarding depression in freezing point?
(a) Vapour pressure of solution is less than that of pure solvent
(b) Vapour pressure of solution is more than that of pure solvent
(c) Only solute molecules solidify at the freezing point.
(d) Only solvent molecules solidify at the freezing point.
Correct are :-

(1) a, d

(2) c, a

(3) b, d

(4) b, c

37) ***Assertion :-*** After adding non-volatile solute freezing point of solvent depressed.

Reason :- Vapour pressure of solvent decreases and become equal to solid solvent, at temperature lower than the freezing point of pure solvent.

(1) Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Assertion and Reason are false

38) **Assertion :** On mixing of two liquids, the solutions becomes hot, it implies that it shows negative deviation from Raoult's Law.
Reason : Solutions which show negative deviations are accompained by increase in volume.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

39) **Assertion :** The concentration of pollutants in water or atmosphere is often expressed in ppm.
Reason : Concentration in parts per million can be expressed as mass to mass, volume to volume and mass to volume.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

40) Match the solute and solvent in column-I with examples in column-II and mark the appropriate choice :-

	Column-I		**Column-II**
(A)	Liquid in gas	(i)	Copper dissolved in gold
(B)	solid in solid	(ii)	Glucose dissolved in water
(C)	Liquid in solid	(iii)	Chloroform mixed with nitrogen gas
(D)	Solid in liquid	(iv)	Amalgam of mercury with sodium

(1) (A)→(i), (B)→(iii), (C)→(ii), (D)→(iv)

(2) (A)→(iii), (B)→(i), (C)→(iv), (D)→(ii)

(3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)

(4) (A)→(iii), (B)→(ii), (C)→(i), (D)→(iv)

41) Select the correctly matched:

(i)	Dalton's law	(P)	$P_T = P_A^0 X_A^0 + P_B^0 X_B^0$
(ii)	Raoult's law (liquid-liquid system)	(Q)	Partial pressure of gas = mole fraction × P_T
(iii)	Raoult's law (solid-liquid system)	(R)	$\frac{P^o - P_S}{P^o} = \frac{n(solute)}{n + N(solvent)}$

(1) (i)-(P), (ii)-(Q), (iii)-(R)

(2) (i)-(P), (ii)-(R), (iii)-(Q)

(3) (i)-(Q), (ii)-(P), (iii)-(R)

(4) (i)-(Q), (ii)-(R), (iii)-(P)

42) Match List-I (solutes) with List-II (ratio of depression in freezing point) and select the correct answer from the codes given below :

	List-I [Solute 0.01 m]		**List-II [ratio of depression in f.p.]**
(A)	Urea, NaCl, Na_2SO_4	(a)	1:2:1
(B)	KCl, NaCl, K_2SO_4	(b)	1:2:3
(C)	Urea, Sugar, Glucose	(c)	2:2:3
(D)	Urea, KCl, Glucose	(d)	1:1:1

Code is :-

(1) A-b, B-c, C-d, D-a

(2) A-a, B-b, C-c, D-d

(3) A-b, B-a, C-c, D-d

(4) A-c, B-d, C-b, D-a

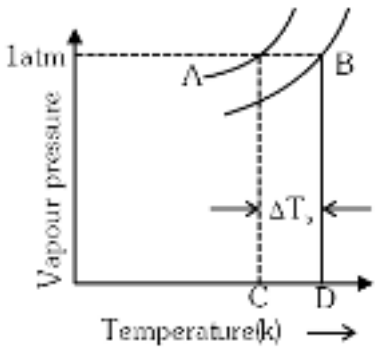
43) Osmosis is the spontaneous net flow through SPM of :-

(1) The solute from a less concentrated solution into more conc. solution.

(2) The solvent from a solution of lower concentration to one of higher concentration

(3) Solute particles from hypotonic solution to hypertonic solution.

(4) Solvent molecules from solution to pure solvent.



44)

The correct match related to above graph is –

(1) (A-solvent) (B-solution) (C-T_b), (D-$T_b°$)

(2) (A-solvent) (B-solution) (C-$T_b°$), (D-T_b)

(3) (A-solution) (B-solvent) (C-$T_b°$), (D-T_b)

(4) (A-solution) (B-solvent) (C-T_b), (D-$T_b°$)

45) Solution of Acetone in ethanol .....

(1) shows positive deviation from Raoult.s law.

(2) acts as an Ideal solution.

(3) Obeys Raoult.s law.

(4) shows negative deviation from Raoult.s law.

BIOLOGY

1) When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of enzyme, it is known as :

(1) Non-competitive inhibitor

(2) Cofactor

(3) Competitive inhibitor

(4) Coenzyme

2) Co-factor are :-

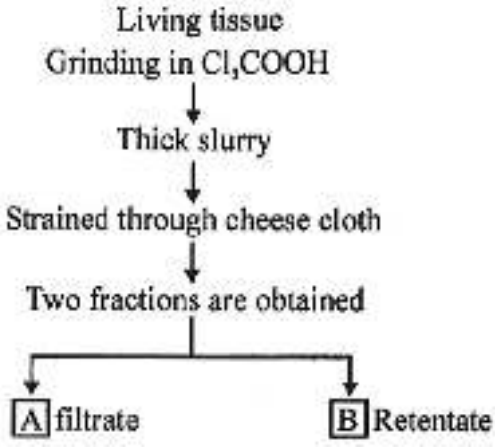
(1) Non-protein part of enzyme

(2) Prosthetic groups, co-enzymes and metal ions

(3) Both (1) and (2)

(4) A specific chemical that always bind to active site enzyme

3) Read the given statements and select the option that correctly sorts these with respect to A and B in the given flow chart :



(a) Acid soluble pool
(b) Acid insoluble pool
(c) Molecular weight ranging from 18 to 800 dalton approximately
(d) Amino acid, sugar, nucleotides
(e) Contain chemicals that have molecular weight more than 1000 D
(f) Protein, nucleic acid, polysaccharides

(1) A - b,e,f ; B-a,c,d

(2) A - a,c,d ; B-b,e,f

(3) A - a,d ; B-b,c,e,f

(4) A - a,c,f ; B-b,e,d

4) Given below are two statements :
Statement I : In a polysaccharide chain, the right end is called the non reducing end and the left end is called the reducing end.
Statement II : Starch forms helical secondary structures into which I_2 molecules trapped.
In the light of above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are false

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are true

5) Match the column I with column II

	Column-I		**Column-II**
(A)	RuBisCO	(I)	Most abundant protein in Animal world
(B)	Insulin	(II)	Enzyme
(C)	Collagen	(III)	Most abundant protein in the whole Biosphere
(D)	Trypsin	(IV)	Hormone
		(V)	Intercellular ground substance

(1) A-III, B-I, C-IV, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-I, C-V, D-II

6) Match **List-I** with **List-II**.

	List-I		**List-II**
A.	Primary structure of protein	I.	Human haemoglobin
B.	Secondary structure of protein	II.	Disulphide bonds
C.	Tertiary structure of protein	III.	Linear chain of amino acid
D.	Quaternary structure of protein	IV.	Alpha helix and β sheet

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-I, D -IV

(3) A-I, B-III, C-II, D-IV

(4) A-IV, B-III, C-II, D-I

7) **Statement-I :-** Many secondary metabolites are useful to human welfare (eg. rubber, drugs, spices, scents and pigments).
Statement-II :- Some secondary metabolites have ecological importance.
Choose the correct answer from the option given below.

(1) Statement I is correct & Statement II is incorrect

(2) Statement I is incorrect & Statement II is correct

(3) Both Statement are correct

(4) Both Statement are incorrect

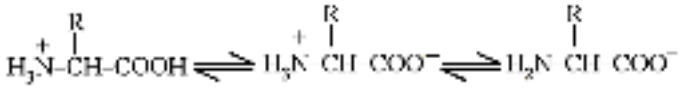
8) Cellulose is a linear polymer of___(A)___units. It has ___(B)___linkage.

(1) (A) β-D-glucose (B) β 1'–4"

(2) (A) β-D-fructose (B) β 1'–4"

(3) (A) α-D-glucose (B) α 1'–4"

(4) (A) β-D-glucose (B) α 1'–6"



9) (A) (B) (C)
Which of the following structure is in form of zwitterion :

(1) Only A

(2) Only B

(3) Only C

(4) Both (A) and (C)

10) In anomers of glucose, position of –H and –OH groups are changed on _____ atom.

(1) C_4

(2) C_3

(3) C_1

(4) C_2

11) Carefully observe the table and select the correct combination :-

	Column-I (process)	**Column-II (Feature/Location)**	**Column-III (Result)**	**Column-IV (Example)**
1.	Cyclic photo phosphorylation	A-Mesophyll cell	X-NADPH and ATP formation	i-Maize
2.	Non-Cyclic photo phosphorylation	B-Stroma lamellae	Y-only ATP formation	ii- C_3 plant

3.	C_4-Pathway	C-Grana Thylakoid	Z-First Stable product is OAA	iii-Both Photo system I & II
4.	C_3- Pathway	D-Mesophyl cell + Bundle sheath cell	W- First stable product is 3-PGA	iv-Photosystem-I only

(1) 1-B-Y-iv, 2-C-X-iii, 3-D-Z-i, 4-A-W-ii

(2) 1-A-X-iii, 2-B-Y-iv, 3-C-Z-ii, 4-D-W-i

(3) 1-C-Z-i, 2-A-W-iv, 3-B-X-iii, 4-D-Y-i

(4) 1-B-Y-ii, 2-C-X-iv, 3-D-Z-iii, 4-A-W-i

12) Given below are two statements :
Statement-I : Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase. Energy is used to pump protons across a membrane to create a gradient or a high concentration of protons with in the thylakoid lumen.
Statement-II : ATP synthase has a channel that allows diffusion of protons back across the membrane. This releases enough energy to activate ATP synthase enzyme that catalyses the formation of ATP.
In the light of the above statements, choose the most appropriate answer from the options given below :

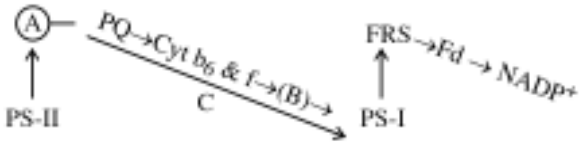
(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

13) Observe the electron flow of Z-scheme and identify missing labelling :



	A	B	C
(1)	Ferredoxin	Plastocyanin	Cytochrome-C
(2)	Phaeophytin	Downhill flow	Ferredoxin
(3)	Plastoquinone	Cytochrome-C	uphill flow
(4)	Phaeophytin	Plastocyanin	downhill flow

(1) 1

(2) 2

(3) 3

(4) 4

14) Identify the correct statements regarding calvin cycle :-
(a) Regeneration is most crucial step of calvin cycle.
(b) Carboxylation is catalysed by the enzyme RuBP carboxylase.
(c) Regeneration steps require 1 ATP for phosphorylation to form one RUBP.
(d) First product of CO_2 fixation is a C_3 acid.

(e) The primary acceptor is a 2 - carbon compound.
Choose the correct answer from the options given below :

(1) (a), (b) and (e)

(2) (c), (d) and (e)

(3) (b), (c) and (d)

(4) (b), (c) and (e)

15) Read the following statement :-
(A) PEPcase has only carboxylase activity.
(B) PEPcase found in both mesophyll and bundle sheath cells of C_4 plants.
(C) RuBisCO has greater affinity for CO_2 than O_2.
(D) PPDK works better at high temperature.
Select the correct option :-

(1) A, B, C only

(2) B, C, D only

(3) A, B, C, D

(4) A, C, D only

16) How many ATPs are required for 3-CO_2 fixation by rice and maize respectively ?

(1) 9, 9

(2) 9, 15

(3) 15, 9

(4) 15, 15

17) Which of the following are the characters of C_4 plants ?
(a) Kranz anatomy
(b) Tolerate higher temperature
(c) RuBisCO present in mesophyll cell
(d) Response to high light intensities
(e) Lack photorespiration
(f) Greater productivity of biomass
How many are correct ?

(1) a, b, c, d are correct

(2) Only c is wrong

(3) all are correct

(4) only d and e are correct

18) Select the incorrect match :-

(1)	Moll's half-leaf experiment	Show that light is required for photosynthesis
(2)	Ruben and Kamen	O_2 in photosynthesis released from water

(3)	Jan ingenhousz	Showed that it is only green part of the plants that could release oxygen
(4)	J.V. Sachs	Provide evidence for production of glucose when plants grow

(1) 1

(2) 2

(3) 3

(4) 4

19) The pigments are organised into ____a____ discrete photochemical ____b____ with in the PS I and PS II :-

(1) a - Four, b - Protein harvesting complexes

(2) a - Three, b- Water harvesting complexes

(3) a - Two, b - Light harvesting complexes

(4) a - Three, b - Air harvesting complexes

20) **Statement-I :-** There is a clear division of labour with in the chloroplast.
Statement-II :- There is a complete one to one overlap between the absorption spectrum of chlorophyll 'a' and action spectrum of photosynthesis.

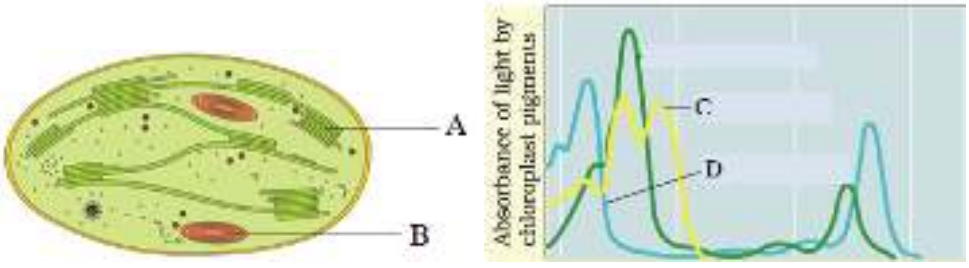
(1) Both statement-I and statement-II are false.

(2) Statement-I is true but statement-II is false

(3) Statement-I is false but statement-II is true.

(4) Both statement-I and statement-II are true.

21) Identify labelled A, B, C & D in diagram and graph given below :-



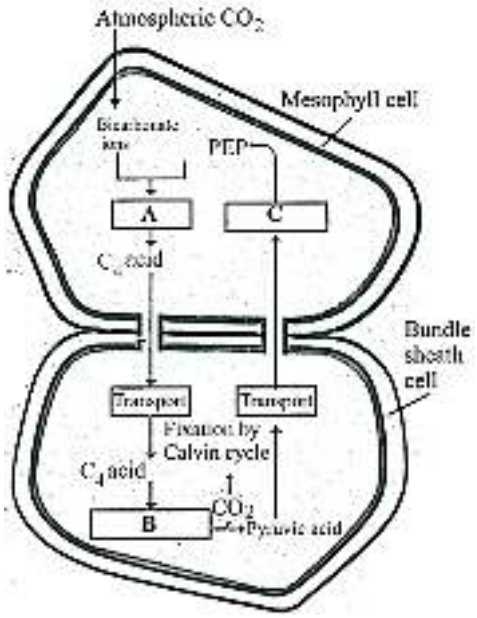
(1) A-Stroma, B-Lipid droplet, C-Chlorophyll a, D-Carotenoids

(2) A-Ribosomes, B-Stroma lamella, C-Chlorophyll b, D-Carotenoids

(3) A-Grana, B-Starch granule, C-Carotenoids, D-Chlorophyll a

(4) A-Stroma lamella, B-Ribosomes, C-Carotenoids, D-Chlorophyll a

22) Given figure represents C_4 pathway. Select the suitable options for A, B and C :-



	A	B	C
(1)	Decarboxylation	Reduction	Regeneration
(2)	Fixation	Transamination	Regeneration
(3)	Carboxylation	Decarboxylation	Reduction
(4)	Fixation	Decarboxylation	Regeneration

(1) 1

(2) 2

(3) 3

(4) 4

23) The electrons needed to replace those removed from photosystem I are provided by :

(1) $NADPH + H^+$

(2) Ferridoxin

(3) PS II

(4) H_2O

24) Which statement is incorrect:

(1) The C_3 and C_4 plants respond similarly to CO_2 concentration

(2) At low light conditions, neither C_3 nor C_4 plants respond to high CO_2 concentration

(3) C_4 plants attain saturation at much lower CO_2 concentration than C_3 plants

(4) Current availability of CO_2 levels is limiting to the C_3 plants

25) How many statement are correct ?
(I) Light saturation occur at 10% full sunlight.
(II) Carbon dioxide is major limiting factor for photosynthesis.
(III) C_4 plant show saturation at about 360 μL^{-1} CO_2.

(IV) Atmospheric CO_2 is not limiting factor for C_4 plant and submerged hydrophytes.

(1) Two
(2) Three
(3) One
(4) Four

26) How many molecules of ATP are required for phosphorylation of 1 molecule of phosphoglycerate in the reduction step.

(1) 2 ATP
(2) 1 ATP
(3) 3 ATP
(4) 4 ATP

27)

Which of the following are **correct** about cellular respiration ?
A. Cellular respiration is the breaking of C-C bonds of complex organic molecules by oxidation.
B. The entire cellular respiration takes place in Mitochondria.
C. Fermentation takes place under anaerobic condition in germinating seeds.
D. The fate of pyruvate formed during glycolysis depends on the type of organism also.
E. Water is formed during respiration as a result of O_2 accepting electrons and getting reduced
Choose the **correct** answer from the options given below :

(1) A, C, D, E only
(2) A, B, E only
(3) A, B, C, E only
(4) B, C, D, E only

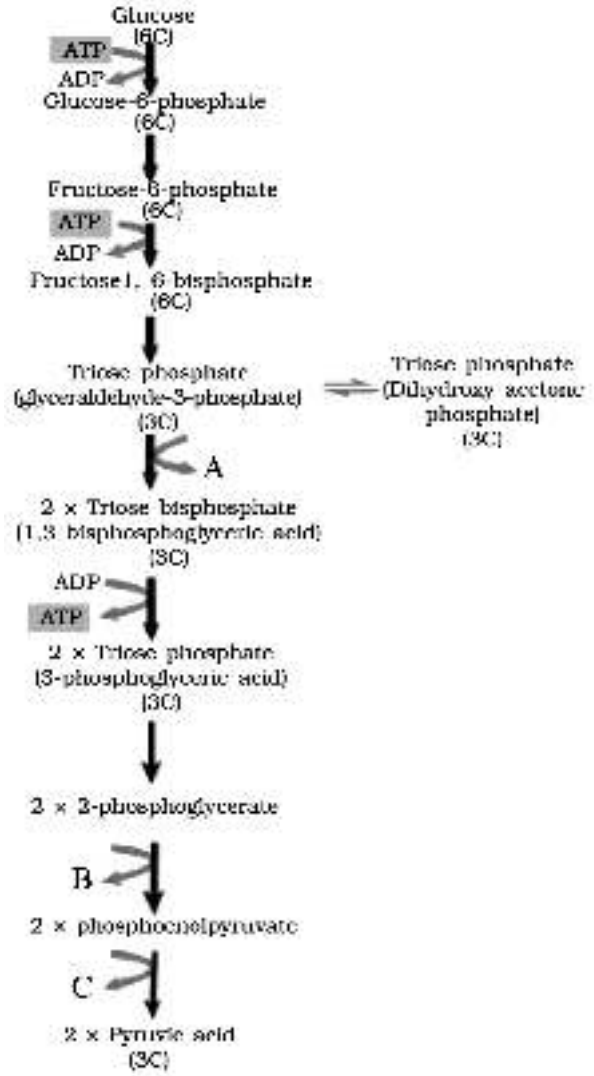
28)

Hexokinase, Pyruvate kinase, Enolase, Mutase, PFK, Pyruvate dehydrogenase, Aconitase, SDH, Malate dehydrogenase, Decarboxylase

How many enzyme given are involved in glycolytic pathway ?

(1) 10
(2) 5
(3) 7
(4) 8

29) Identify A, B and C in the given diagram of glycolysis :-



	A	**B**	**C**
1	ATP	ATP	H_2O
2	$NADH+H^+$	H_2O	ATP
3	$NADH+H^+$	ATP	ATP
4	H_2O	ATP	ATP

(1) 1

(2) 2

(3) 3

(4) 4

30) $2(A) + 145O_2 \rightarrow 102\ CO_2 + nH_2O$ + Energy

In given reaction choose the correct match for A and n in given options.

(1) A-$C_{51}H_{98}O_6$, n-96

(2) A-$C_{51}H_{98}O_5$, n-98

(3) A-$C_{50}H_{98}O_6$, n-96

(4) A-$C_{51}H_{98}O_6$, n-98

31) Although the aerobic process of respiration takes place only in the presence of oxygen, the role of oxygen is limited to the ____A____ of the process. yet, the presence of oxygen is vital, since it drives the whole process by ____B___ from the system

(1) A = ETS, B = removing electron

(2) A = Krebs cycle, B = Removing hydrogen

(3) A = Terminal stage, B = Removing hydrogen

(4) A = Link reaction and Krebs cycle, B = Removing electron

32) In aerobic respiration, one glucose produces :-

(1) 8 NADH + 3 $FADH_2$ + 3 ATP

(2) 12 NADH + 2 $FADH_2$ + 38 ATP

(3) 12 NADH + 30 ATP + H_2O

(4) 10 NADH + 2 $FADH_2$ + 2ATP + 2 GTP

33) In which of the following respiratory process decarboxylation does not takes place ?

(1) Pyruvate → Acetyl CoA

(2) Acetyl CoA → Malic acid

(3) Pyruvate → Lactic acid

(4) Pyruvate → Ethyl alcohol

34) The donor and acceptor of e^- & H^+ in alcoholic fermentation are :-

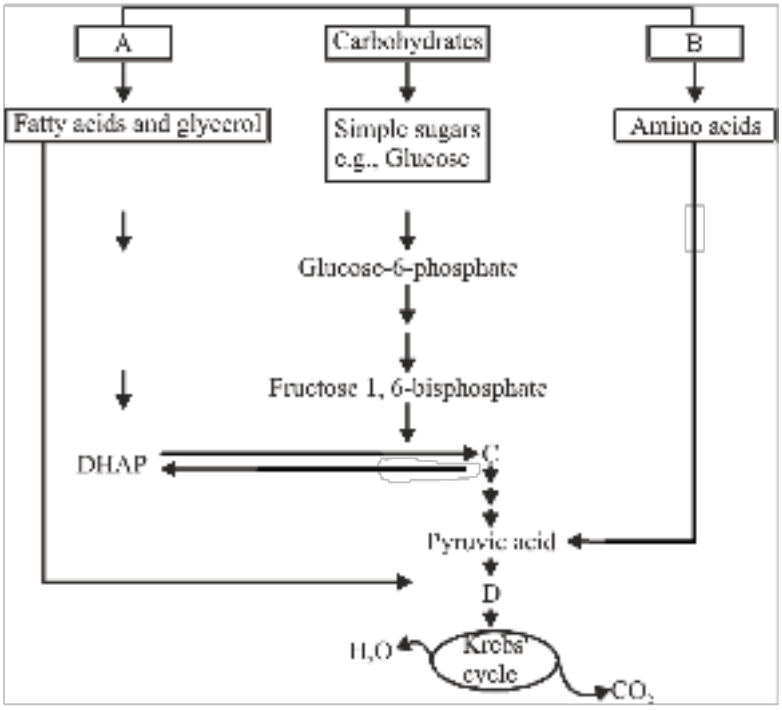
(1) Glucose and pyruvate respectively

(2) DHAP and pyruvate respectively

(3) DHAP and acetaldehyde respectively

(4) PGAL and acetaldehyde respectively

35) Select the correct option for A, B, C and D :-



	A	B	C	D
(1)	Fats	Proteins	3-PGAL	Acetyl CoA
(2)	Fats	Proteins	3-PGAL	CO_2
(3)	Proteins	Fats	Acetyl CoA	PEP
(4)	Proteins	Fats	PEP	Acetyl CoA

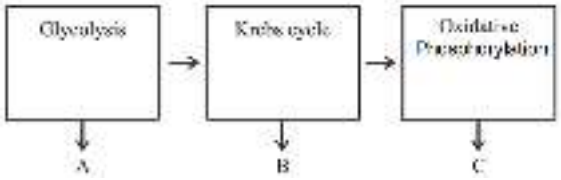
(1) 1

(2) 2

(3) 3

(4) 4

36)



If one molecule of glucose involve in respiration then how many ATP are gained by substrate level phosphorylation respectively in A, B and C process :

(1) A–2, B–2, C–2

(2) A–2, B–2, C–0

(3) A–4, B–0, C–4

(4) A–4, B–0, C–2

37) (I) Proton channel of oxysome/complex-V/ATP synthase is F_0.
(II) Metabolic water is water produced in terminal oxidation in respiration
(III) UQ accepts electron from NADH dehydrogenase (complex I) and also can accept electron from $FADH_2$ complex II.

(IV) Cytochrome c is a small protein attached to outer surface of the inner mitochondrial membrane and acts as a mobile carrier for transfer of electrons between complex III and complex IV.
(V) Complex IV refers to cytochrome C oxidase.
Choose the corrects option :–

(1) All are correct
(2) All are incorrect
(3) Only I and V are correct
(4) Only III is correct

38) Pyruvate dehydrogenase complex, responsible for pyruvate decarboxylation in cellular respiration in eukaryotes, is found in :-

(1) Perimitochondrial space
(2) Mitochondrial matrix
(3) Inner mitochondrial membrane
(4) Cytoplasm

39) Conversion of fructose-6-phosphate to fructose-1, 6 bisphosphate, the second irreversible reaction of glycolysis, is catalysed by:

(1) Aldolase
(2) Hexokinase
(3) Phosphofructokinase
(4) Enolase

40) (A) Pyruvic acid, change to acetyl CoA before entering in TCA-cycle.
(B) Succinyl CoA is only five carbon intermediate in TCA-cycle.
(C) TCA cycle occurs in matrix of mitochondria.
(D) Glycolysis occurs in cytosol.
Choose the option which have all the correct statements from A to D :-

(1) A, B, C
(2) A, B, D
(3) B, C, D
(4) A, C, D

41) Respiratory pathway has traditionally been considered as __X__ but it would be better to considered as __Y__. X & Y are respectively-

(1) Amphibolic pathway, Anabolic pathway
(2) Anabolic pathway, Catabolic pathway
(3) Catabolic pathway, Amphibolic pathway
(4) Amphibolic pathway, Catabolic pathway

42) The complete oxidation of pyruvate by the stepwise removel of all the hydrogen atoms takes place in the -

(1) Cytoplasm

(2) Matrix of mitochondria

(3) Inner membrane of mitochondria

(4) Inter membrane space of mitochondria

43) Match the following columns :

	Column-I		Column-II		Column-III
(A)	Water splitting complex	P	Herbicide	1	Evolution of oxygen
(B)	Tricarboxylic acid cycle	Q	Gaseous hormone	2	Weed free lawn
(C)	Auxin	R	PS-II	3	Krebs cycle
(D)	Ethylene	S	Aerobic respiration	4	Petiole elongation in deep water rice plant

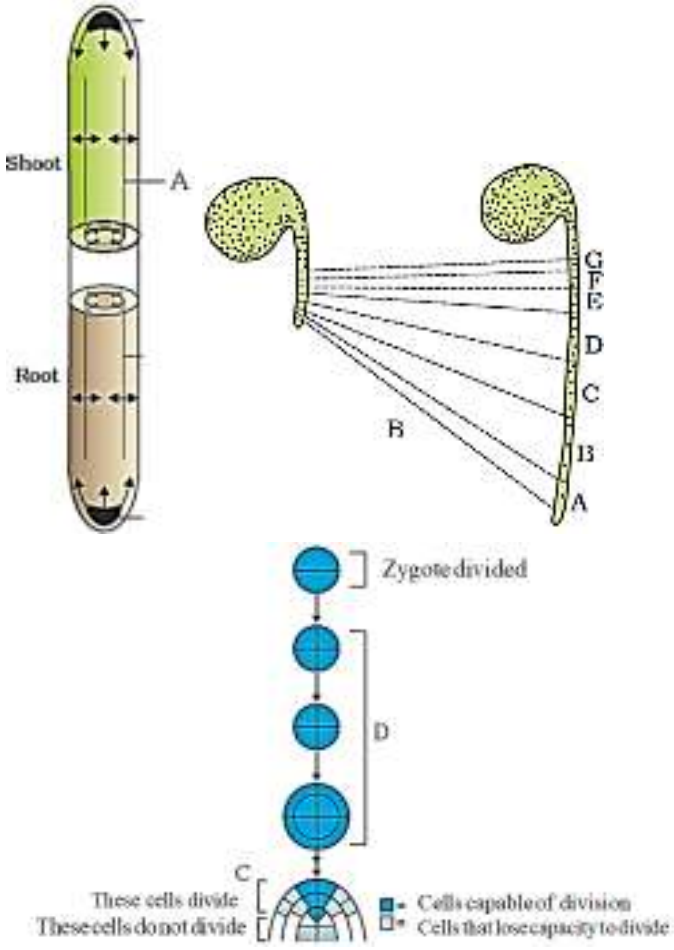
(1) A-R-1, B-S-2, C-P-3, D-Q-4

(2) A-R-1, B-S-3, C-P-2, D-Q-4

(3) A-R-1, B-S-3, C-P-4, D-Q-2

(4) A-R-1, B-S-2, C-P-4, D-Q-3

44) Find the labelled A, B & C & D in given figure.



	A	B	C	D
(1)	Vascular cambium	Parallel line technique	Geometric phase	Arithmetic phase
(2)	Cork cambium	Parallel line technique	Geometric phase	Arithmetic phase
(3)	Vascular cambium	Parallel line technique	Arithmetic phase	Geometric phase
(4)	Cork cambium	Relative growth	Arithmetic phase	Geometric phase

(1) 1

(2) 2

(3) 3

(4) 4

45) Read the following terminology of column-I & II & identify their correct match :

	Column-I		Column-II
(A)	Abscisic acid	(i)	Promotes female flower in cucumber
(B)	Auxin	(ii)	Promotes bolting in beet & cabbage
(C)	Cytokinin	(iii)	Induce parthenocarpy in tomato
(D)	Gibberelins	(iv)	Stimulates the closure of stomata
		(v)	Promotes root growth & root hair formation
		(vi)	Promotes nutrient mobilisation

(1) A-iii, B-v, C-vi, D-ii

(2) A-iv, B-iii, C-v, D-ii

(3) A-iv, B-iii, C-vi, D-ii

(4) A-i, B-iv, C-iii, D-v

46) Match column-I with column-II and select the correct option from the codes given below :-

	Column-I		Column-II
A	C.Darwin and F.Darwin	i	Cytokinin
B	Miller and Skoog	ii	GA
C	H. H. Cousins	iii	Ethylene
D	Kurosawa	iv	Auxin

(1) A-(iv), B-(i), C-(iii), D-(ii)

(2) A-(iv), B-(i), C-(ii), D-(iii)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(ii), D-(i)

47) **Assertion (A) :-** Abscisic acid is known as stress hormone.
Reason (R) :- Abscisic acid stimulates the closure of stomata and increase the tolerance of plant to various kinds of stresses.

(1) (A) & (R) both are correct and (R) is correct explanation of (A).

(2) (A) & (R) both are correct but (R) is not correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) (A) and (R) both are incorrect

48) Pick out the correct statements
(a) Cytokinins specially help in delaying leaf senescence
(b) ABA are involved in regulating apical dominance
(c) Ethylene is specially useful in promoting fruit ripening
(d) Gibberellins are responsible for immature falling of leaves

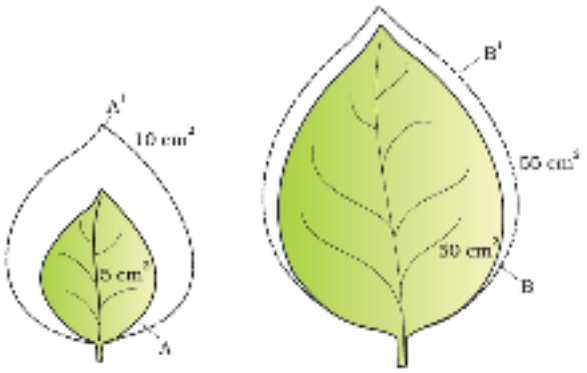
(1) a and c only

(2) a and d only

(3) b and c only

(4) a and b only

49)



Find the correct one with respect to above diagram :-

(1) Absolute growth rate in leaf A & leaf B is different

(2) Relative growth rate in leaf B is more than leaf A

(3) Despite 100% relative growth rate in leaf A, leaf B is large in size than leaf A

(4) Relative growth rate in leaf A and leaf B is equal.

50) **Assertion :-** Plant growth is unique because plants retain the capacity for unlimited growth throughout their life. **Reason :-** The product of meristem loses the capacity to divide and such cells which lost the division capacity make up the plant body.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

51) (I) O_2 helps in releasing metabolic energy essential for growth activities

(II) Nutrients (micro and macroelements) are required by plants for the synthesis of protoplasm as well as act as source of energy
(III) Optimum temperature for plant growth is needed
(IV) Environmental signal like light and gravity affect certain phases of growth
Choose the correct option

(1) All are correct
(2) Only I, II and III are correct
(3) Only I, II and IV are correct
(4) Only I and IV are correct

52) Cells regain the capacity of division under certain condition called as :-

(1) Differentiation
(2) Quiescence
(3) Dedifferentiation
(4) Redifferentiation

53) In sigmoid growth curve, the exponential phase slows down leading to a stationary phase due to :-

(1) Toxic effect of some plant hormones
(2) Genetic nature
(3) Limited nutrient supply
(4) Absence of differentiation.

54) The discovery of which plant hormone is related to D.N.A. :-

(1) Cytokinins
(2) Auxins
(3) Gibberellins
(4) Ethylene

55) What is the correct sequence of electron carriers during e^- transfer in ETS ?

(1) $NADH_2 \rightarrow$ Cyt c $\rightarrow$ UQ $\rightarrow$ Cyt b
(2) $NADH_2 \rightarrow$ FMN $\rightarrow$ UQ $\rightarrow$ Cyt b
(3) $FADH_2 \rightarrow$ Cyt c $\rightarrow$ Cyt b $\rightarrow$ Cyt c_1
(4) Cyt c $\rightarrow$ Cyt $c_1 \rightarrow$ Cyt b $\rightarrow$ Cyt a_3

56) Math List-I with List-II-

	List-I		**List-II**
(A)	NAD & NADP	(I)	Carboxypeptidase
(B)	Zn	(II)	Haem

(C)	Peroxidase	(III)	Coenzyme
(D)	Succinic dehydrogenase	(IV)	Malonate

Choose the correct answer from the options given below-

(1) (A–III), (B–IV), (C–I), (D–II)

(2) (A–III), (B–I), (C–IV), (D–II)

(3) (A–III), (B–I), (C–II), (D–IV)

(4) (A–II), (B–I), (C–III), (D–IV)

57) **Statement I:** Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds are called Lyases.
Statement II: Enzymes that catalyse interconversion of optical geometric or positional isomers are called Hydrolases.

(1) Statement I and Statement II are correct and Statement II explains Statement I

(2) Statement I and Statement II are correct but Statement II does not explain Statement I

(3) Statement I is correct but Statement II is incorrect

(4) Both Statements are wrong

58) The catalytic cycle of an enzyme action is described as :
(A) Substrate binds to the active site of the enzyme.
(B) The active site now in close proximity of the substrate breaks the chemical bonds of the substrate.
(C) The binding of the substrate induces the enzymes to alter its shape.
(D) Active site fits more tightly around the substrate.
(E) Enzyme releases the products of the reaction.
Choose the correct answer from the options given below :

(1) A → C → B → D → E

(2) A → C → D → B → E

(3) A → B → C → D → E

(4) B → A → D → C → E

59)

Which of the following is not a factor for photorespiration ?

(1) High CO_2 concentration

(2) High light intensity

(3) High temperature

(4) High O_2 concentration

60) (A) Catalyses the transfer of a group (other than hydrogen)
(B) Catalyses removal of group from substrates by mechanism other than hydrolysis leaving double bond
(C) Catalyses the linking of two molecules by using energy into one molecule
Identify the classes of enzymes for the above function :-

	A	B	C
(1)	Ligase	Lyases	Transferase
(2)	Transferase	Lyases	Ligase
(3)	Lyase	Ligase	Transferase
(4)	Transferase	Ligase	Lyase

(1) 1

(2) 2

(3) 3

(4) 4

61) The stroma of chloroplast has :-

(1) Enzyme for protein synthesis

(2) Enzyme for carbohydrate synthesis

(3) Both (1) and (2)

(4) Enzyme for dark reaction and oxidative phosphorylation

62) Plant cells and animal cells are different as the former possess

(1) Mitochondria, cytoplasm and golgi apparatus

(2) Cell wall, plastids and a large central vacuole

(3) Centrioles, microtubules and microvilli

(4) Peroxisome, lysosome and nucleus

63) **Assertion (A):** Peripheral microtubules of a flagellum are structurally different from the central microtubules.
Reason (R): Peripheral microtubules are doublets, whereas central microtubules are singlets enclosed by a central sheath.

(1) both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

64) Which of the following correctly matches organelle – function?

(1) Lysosome – ATP synthesis

(2) SER – Lipid synthesis

(3) Ribosome – Detoxification

(4) Golgi – DNA replication

65) The primary function of nuclear pore complex is:

(1) DNA replication

(2) Transcription

(3) Selective nucleocytoplasmic transport

(4) Ribosome assembly

66) Select the correct statement from the following regarding cell membrane.

(1) Lipid make up 50% of the erythrocyte cell membrane.
(2) Na^+ and K^+ ion move across cell membrane by passive transport.
(3) Lipids are arranged in a bilayer with polar heads towards the inner part.
(4) Neutral solutes may move across the membrane by simple diffusion.

67) Find the incorrect statement belong to endoplasmic reticulum

(1) Network of tiny tubular structures scattered in the cytoplasm
(2) They provide site to ribosome for protein synthesis
(3) They can be involved in steroid synthesis
(4) They contain active hydrolytic enzymes

68) Match the column-I and Column-II and identify the correct option.

Column-I		**Column-II**	
(A)	Centriole	(i)	Migration of chromosomes towards opposite poles
(B)	Centromere	(ii)	Disintegration mark the start of second phase of mitosis
(C)	Anaphase	(iii)	Chromatids separate
(D)	Nuclear membrane	(iv)	2C convert into 4C
		(v)	9+2 arrangement of microtubules
		(vi)	9+0 arrangement of microtubules
		(vii)	Hold two chromatids of chromosomes

(1) A-v ; B-vii ; C-ii, iv ; D-ii
(2) A-vi ; B-ii ; C-i, iv ; D-iii
(3) A-vi ; B-vii ; C-i, iii ; D-ii
(4) A-i, vii ; B-i ; C-ii, iv ; D-ii

69) Select the number of correct statements from the following :
I. Cell-plate formation occurs in plant cell during cytokinesis.
II. During cytokinesis mitochondria and plastids get distributed between two daughter cells in mitosis.
III. Liquid endosperm in coconut is syncytium.
IV. Furrow formation occurs in animal cell during cytokinesis.

(1) 1
(2) 2
(3) 3

(4) 4

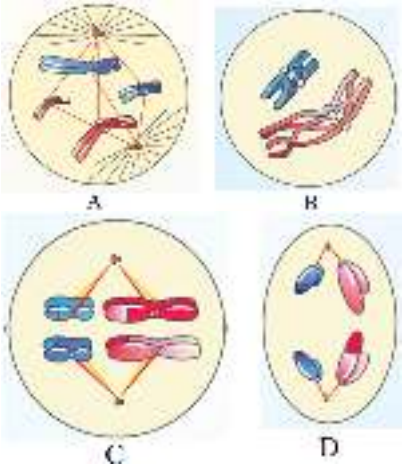
70) What will be the number of chromatids in prophase-I of a cell having 20 bivalents?

(1) 20
(2) 80
(3) 40
(4) 10

71) In human cells during meiosis how many cycles of nuclear division, DNA replication and cell division occurs respectively ?

(1) One, One, Two
(2) Two, Two, One
(3) One, Two, One
(4) Two, One, Two

72) Identify the figures given below and choose the **correct** options :



(1) A-Transition to metaphase, B-Prophase, C-Anaphase-II, D-metaphase-II
(2) A-Transition to metaphase, B-prophase-II, C-Metaphase-I, D-Anaphase-II
(3) A-Transition to metaphase, B-Telophase-I, C-Metaphase, D-Anaphase-I
(4) A-Transition to metaphase, B-Prophase-I, C-Metaphase-I, D-Anaphase-II

73) Mitotic spindle is mainly composed of which proteins ?

(1) Tubulin
(2) Myosin
(3) Actomyosin
(4) Myoglobin

74) **Statement-I :** Synaptonemal complex is a structure involved in pairing of homologous

chromosome.
Statement-II : Crossing over occurs between non-sister chromatids of homologous chromosomes.

(1) Only statement I is correct.

(2) Only statement II is correct.

(3) Both statements are correct.

(4) Both statements are incorrect.

75) Which is most dramatic period of cell cycle.

(1) G_1 - phase

(2) G_2 - phase

(3) S - phase

(4) M - phase

76) Given below are two statements :-
Statement-I : In cymose type of inflorescence the main axis terminates in a flower, hence is limited in growth. The flowers are borne in a basipetal order.
Statement-II : When shoot tip transforms into a flower, it is always solitary.
In the light of the above statement, choose the correct answer from the option given below

(1) Both statements-I and statement-II are incorrect

(2) Statement-I is correct but statement-II is incorrect

(3) Statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

77) Identify the incorrect statement regarding leaf :-

(1) Leaf originate from shoot apical meristem and are arranged in acropetal order .

(2) In leaves of all plants, petiole become swollen up which is called pulvinus.

(3) The leaf is attached to the stem by the leaf base and may bear two lateral small leaf like structures called stipules.

(4) Leaves are the most important vegetative organ for photosynthesis.

78) Match the following column A with B -

Column - A		**Column - B**	
(i)	Marginal	(a)	Lemon
(ii)	Axile	(b)	Pea
(iii)	Free central	(c)	Marigold
(iv)	Basal	(d)	*Dianthus*

(1) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)

(2) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

(3) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)

(4) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)

79) Select the correct statement ?

(1) Mesocarp of mango is fleshy and coconut is fibrous.
(2) Mesocarp of coconut is fleshy and mango is fibrous.
(3) Endocarp of coconut is stony and mango is fleshy.
(4) Endocarp of mango is fleshy and coconut is fibrous.

80) Actinomorphic or radial symmetry is shown by ?

(1) Mustard, Pea, Bean
(2) Gulmohar, *Casia*, Mustard
(3) *Datura*, Pea, *Canna*
(4) Mustard, Chilli, *Datura*

81) What type of phyllotaxy is found in guava and chinarose respectively ?

(1) Alternate, alternate
(2) Opposite, opposite
(3) Opposite, alternate
(4) Opposite, whorled

82) From the options given below, find out the correct floral formula for a flower having the following characters namely actinomorphic, bisexual, five united sepals, five united petals, Stamens five free and epipetalous, bicarpellary syncarpous with superior ovary :

(1) $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C_{(5)}\ A_5}\ \underline{G}_{(2)}$

(2) $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C_{(5)}\ A_{(5)}}\ \underline{G}_{(2)}$

(3) $\oplus\ \female\ K_5\, C_5\, A_5\, \underline{G}_{(2)}$

(4) $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C_{(5)}\ A_{(5)}}\ \bar{G}_{(2)}$

83) Match Column I and Column II with Column III.

(A)	Marginal	(I)		(1)	Sunflower, Marigold

(B)	Axile	(II)		(2)	Dianthus, Primrose
(C)	Parietal	(III)		(3)	Mustard, Argemone
(D)	Free Central	(IV)		(4)	China rose, Tomato, Lemon
(E)	Basal	(V)		(5)	Pea

(1) A-V, 5; B-II, 4; C-I, 3; D-III, 2; E-IV, 1

(2) A-I, 5; B-II, 4; C-III, 3; D-IV, 2; E-V, 1

(3) A-V, 1; B-II, 4; C-I, 2; D-III, 3; E-IV, 5

(4) A-V, 1; B-III, 2: C-II, 4: D-I, 5; E-IV, 3

84) Consider the following four statement (a - d) and select the option which include all the correct ones only.
(a) Guard cells are chlorophyllous epidermal cells having outer thin and inner thick walls.
(b) In root, protoxylem lies towards periphery and metaxylem lies the centre.
(c) Phloem fibres are pointed needle like apices found in secondary phloem.
(d) Root hairs are unicellular branched elongation of epidermal cells. **Option :**

(1) a, b and c

(2) a and b

(3) a, c and d

(4) a, b and d

85) Monocot root differs from dicot root in having :

(1) Open vascular bundles

(2) Scattered vascular bundles

(3) Well developed pith

(4) Radially arranged vascular bundles

86) When protoxylem is present towards pericycle and metaxylem is present towards pith this type of primary xylem is known as :-

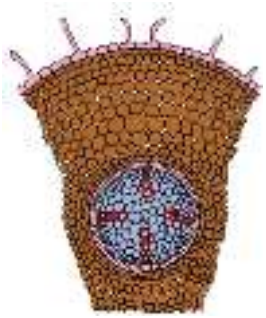
(1) Endarch

(2) Exarch

(3) Mesarch

(4) Radial

87) Study the following diagrams carefully and identify the structure is in which casparian strips is/are present :

A. 

B. 

C. 

D. 

(1) A and C

(2) Only A

(3) A, B and C

(4) A, B, C and D

88) Which of the following is true for bulliform cells?

(1) They are present in abaxial epidermis of grasses

(2) When they are flaccid they increases transpiration

(3) They are large, empty, colourless cell

(4) They are present on both surface of grass leaves

89) Read the statement I and statement II and select the correct option :-
Statement-I :- In dicot stem epidermis is outermost layer and pith is present in peripheral position
Statement-II :- In dicot stem pericycle present above the endodermis and below the cortex Option :-

(1) Statement I and II both are correct

(2) Only statement I is correct

(3) Only statement II is correct

(4) Statement I and II both are incorrect

90) The anatomy of young dicotyledonous stem shows some peculiar features. Identify the correct set of statements about young dicotyledonous stem.
(a) Its hypodermis is sclerenchymatous

(b) Its hypodermis is collenchymatous
(c) Its endodermis is also refferred to as starch sheath
(d) Its pericycle is located just above phloem bundles in the form of semilunar patches of sclerenchyma called bundle cap.
(e) Its vascular bundles are arranged in a ring and conjoint, collated and open with endarch protoxylem.
Choose the correct answer from the options given below :-

(1) (a), (c), (d) and (e) only

(2) (a), (d) and (e) only

(3) (b), (c), (d) and (e) only

(4) (a), (c) and (e) only

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	1	4	4	4	1	3	2	2	1	3	3	3	2	4	4	1	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	4	2	1	4	1	2	4	1	2	2	3	4	1	4	4	4	4	2	2
Q.	41	42	43	44	45															
A.	4	4	4	2	3															

CHEMISTRY

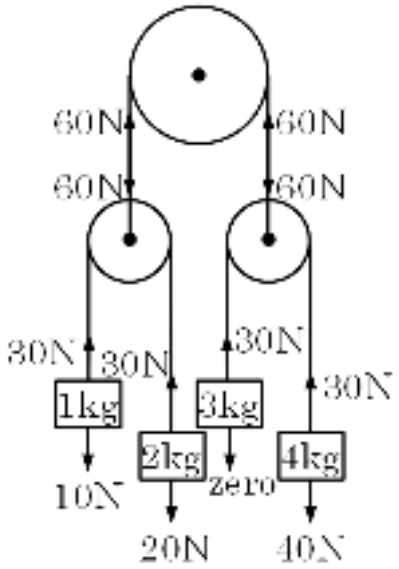
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	4	1	2	2	3	2	4	4	4	4	4	4	3	3	2	3	4	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	3	2	1	3	1	1	4	2	3	1	3	4	1	1	1	3	2	2
Q.	86	87	88	89	90															
A.	3	1	2	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	2	3	2	1	3	1	2	3	1	1	4	3	4	2	2	1	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	1	4	2	1	2	2	4	3	4	3	4	1	2	1	2	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	3	3	1	1	1	3	2	1	3	3	1	2	3	4	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	1	2	3	4	4	3	4	2	4	4	1	3	4	4	2	2	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	1	1	3	2	1	3	4	3										

SOLUTIONS

PHYSICS



1)

On 1 kg; $30 - 10 = 1 \times a_1$

$a_1 = 20\ m/s^2$

On 2 kg; $30 - 20 = 2 \times a_2$

$a_2 = 5\ m/s^2$

On 3 kg; $30 - 30 = 3 \times a_3$

$a_3 = 0$

On 4 kg; $40 - 30 = 4 \times a_4$

$a_4 = 2.5\ m/s^2$

2)

When lift in accelerating upwardN = m(g + a)

when lift in going down & slowing down

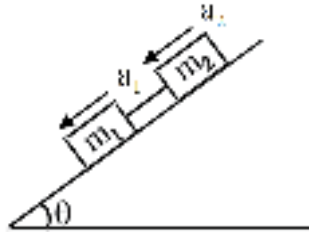
N = m(g + a)

Therefore, the force exerted by the lift is greater than the person's weight when the lift is either going up and speeding up or going down and slowing down.

3)

$T > 0$ if $a_1 > a_2$

$a = g\sin\theta - \mu g\cos\theta$



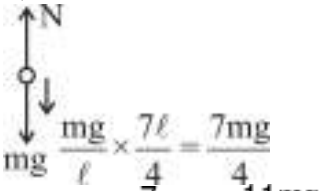
so if μ ↑ a ↓

so T > 0 if $\mu_1 < \mu_2$
otherwise T = 0 for $\mu_1 \geq \mu_2$

$$F = \frac{dP}{dt} = \frac{d}{dt}(4 + 5t^2) = 10t$$

4) $(F)_{t=2} = 10 \times 2 = 20\ \text{N}$

5) Extension = $2\ell - \frac{\ell}{4} = \frac{7\ell}{4}$



$$\frac{mg}{\ell} \times \frac{7\ell}{4} = \frac{7mg}{4}$$

$$N = mg + \frac{7mg}{4} = \frac{11mg}{4}$$

6)

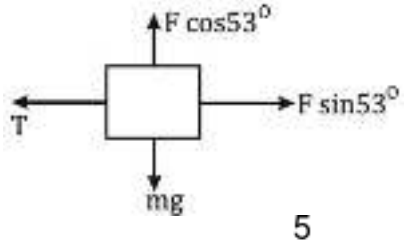
For FBD we draw forces acting on the body



7)

2T + T +T = 60g (man + plank)
4T = 600
T = 150 N

8) FBD of block



Fcos 53° = mg ⇒ F = $\frac{5}{3}$ mg
When spring is cut T = 0
F sin53° = ma

$$a = \frac{F}{m} \cdot \frac{4}{5} = \frac{5}{3} mg \times \frac{4}{5m}$$

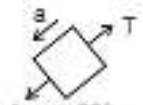
$a = \frac{4g}{3}$

9)

Case - I ⇒ $a = \frac{2F}{4m} = \frac{F}{2m}, N + F = 3ma$

N = 0.5 F

Case - II ⇒ $a_1 = a_2 = \frac{F}{m}$, N = 0

10) 

$4g\sin 60° - T = 4a$ ...(1)



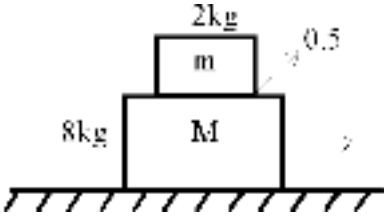
$T - g\sin 30° = a$ ...(2)

Solving (1) and (2) we get.

$20\sqrt{3} - T = 4T - 20$

$T = 4(\sqrt{3}+1)N$

11) For 2 Kg
$T = 2a$
For 3 Kg
$50 - 2T = 3\frac{a}{2}$

12) 

$(a_A)_{max} = 0.5g = 4.9m/s^2$

For moving together



$F_{max} = m_T a_A$

$= 10 \times 4.9$

$= 49N$

13) F = ma

∴ μ g = ma

⇒ $\mu = \frac{a}{g}$

Now, v = u + at or 0 = 6 + 10a

or $\frac{0.6}{10} = a = -0.6$

so $\mu = \frac{a}{g} = \frac{0.6}{10} = -0.6$

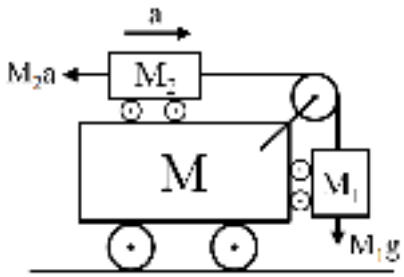
14) u = 36km h^{-1} = 10 ms^{-1},

v = u + at

O = 10 – a × 5 ⇒ a = 2

From $\tan\theta = \frac{a}{g}$, $\tan\theta = \frac{2}{10}$

15)



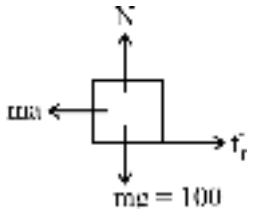
$M_2a = M_1g$

$a = \frac{M_1}{M_2}g$

$F = (M + M_1 + M_2)a$

$=(M + M_1 + M_2)\frac{M_1}{M_2}g$

16)



$f_L = \mu_s N$

$f_L = 0.5 \times 100 \Rightarrow f_L = 50$ N

and ma = 10 × 3 = 30 N

ma < f_L [driving force (ma is less than limiting friction force so block will not move on surface.]

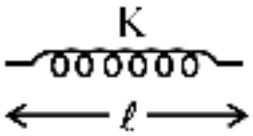
So friction force will be static and equal to f = ma

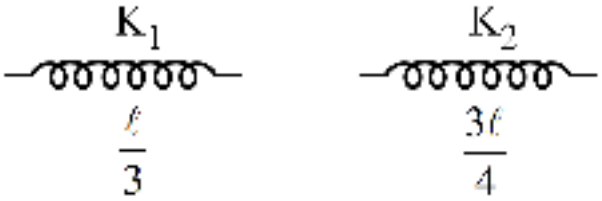
f = 30 newton

17) $2T\cos\left(\frac{\theta}{2}\right) = mg.$

$T \propto \cos\left(\frac{\theta}{2}\right)$

$T \propto \theta$





18)

$K\ell$ = constant

$\therefore K_1\ell_1 = K_2\ell_2 = K\ell$

$$K_2 = \frac{K\ell}{\ell_2} = \frac{K\ell 4}{3\ell}$$

$$K_2 = \frac{4K}{3}$$

19)

$$MP = -\frac{f_0}{fe}\left(1 + \frac{fe}{D}\right)$$

$$= \frac{-100}{10}\left(1 + \frac{10}{25}\right)$$

MP = –14

20)

Solution:

$m = m_0 \times m_e$

$$m = \left[\frac{f_0}{f_0 + u_0}\right] \times \frac{D}{f_e}$$

$$m = \left[\frac{4}{4 - 4.5}\right] \times \frac{24}{8}$$

= 24

Final answer: option (3)

21)

L = 25 cm = $v_0 + f_e$

25 = v_0 + 2.5

v_0 = 22.5 cm

f_0 = 1.5 cm

$$\frac{1}{22.5} - \frac{1}{u_0} = \frac{1}{1.5}$$

$$\frac{1}{u_0} = \frac{1}{22.5} - \frac{1}{1.5} = \frac{10 - 150}{225}$$

$$u_0 = \frac{-225}{140}$$

$$M.P. = \frac{v_0}{u_0} \times \frac{D}{f_e}$$
$$= \frac{22.5 \times 140 \times 25}{225 \times 2.5} = 140$$

22) $MP_{min} = \frac{D}{f} = \frac{25}{12.5} = 2$

$MP_{max} = \frac{D}{f} + 1 = \frac{25}{12.5} + 1 = 3$

23)

For achromatic doublet

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$
$$\Rightarrow \frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2}$$
$$\boxed{\frac{f_2}{f_1} = -1.5} \quad \text{....(1)}$$
$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{90} \quad \text{....(2)}$$

Solving eq.(1) and eq. (2)

$f_1 = 30$ cm ; $f_2 = -45$ cm

24)

For real image, in displacement method

$D \geq 4f \quad \Rightarrow \quad 60 \geq 4f \quad \Rightarrow f \leq 15$ cm

25) **Solution:** Given: u = f + x = 10 + 5 = –15 cm

f = +10 cm

Calculation: $\frac{1}{v} - \frac{1}{-15} = \frac{1}{+10}$

v = 30 cm

26)

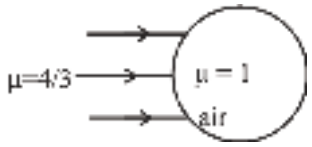
Solution :

$$\frac{1}{2} = (1 \cdot 5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \Rightarrow 1 = \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{f'} = \left(\frac{1 \cdot 5}{1 \cdot 25} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{f'} = \left(\frac{0.25}{1 \cdot 25}\right) \times 1 \Rightarrow f' = 5 \text{ cm}$$

27) $f_1 = 40$ cm, $f_2 = -25$ cm
power of resultant combination
$P = P_1 + P_2 = \frac{100}{40} + \frac{100}{-25}$
$P = 2.5 - 4 = -1.5$ D

28)

$u = mf$
magnification $= \frac{f}{f+u} = \frac{-f}{-f-mf} = \frac{1}{1+m}$

29) 

$\frac{1}{v_1} - \frac{4/3}{\infty} = \frac{1-4/3}{2} = \frac{-1}{6}$

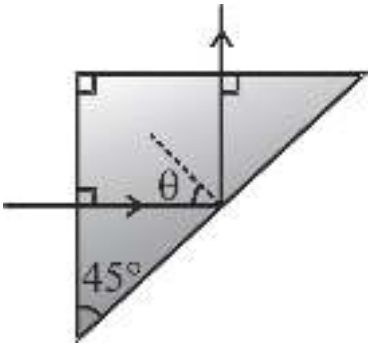
30) 

$1 \times \sin 2A = \mu \times \sin A \Rightarrow \mu = 2\cos A$

31) $\delta = i + e - A$
$\delta = 2i - A$
$\delta = 2\left(\frac{3}{4}A\right) - A$
$\delta = \frac{A}{2}$ ∴ $A = 60°$
$\therefore \delta = 30°$

32)

SOLUTION :



Given internal incidence angle on slanted face = 45°.
For total internal reflection: $45° > i_c$.

Thus $i_c < 45°$.

$$\sin i_c < \sin 45^\circ$$

$$\sin i_c < \frac{1}{\sqrt{2}}$$

Using $\sin i_c = \frac{1}{\mu}$

$$\frac{1}{\mu} < \frac{1}{\sqrt{2}}$$

$\mu > \sqrt{2}$

$\mu > 1.41$.

33)

QUESTION EXPLANATION: A cylindrical glass rod ($\mu = \frac{2}{\sqrt{3}}$) is in air. A ray enters at the center of the flat end at angle θ. Find θ such that the ray inside the rod grazes the curved wall (i.e., hits wall at critical angle).

CONCEPT: Total internal reflection in cylindrical rod.

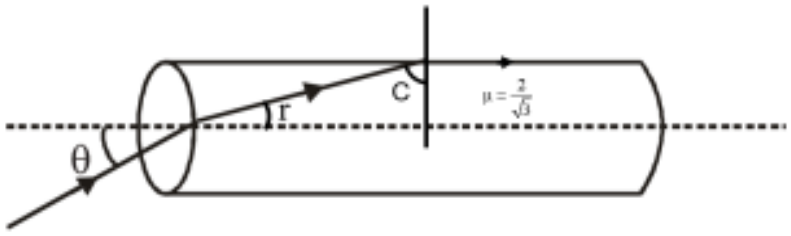
FORMULA:

$$\sin C = \frac{1}{\mu}$$

$i + r = 90°$ for grazing at the cylindrical wall

Snell's law at entrance: $\sin \theta = \mu \sin r$

SOLUTION:



Critical angle at glass–air boundary:

$$\sin C = \frac{1}{\mu} = \frac{\sqrt{3}}{2}$$

Grazing along wall means inside incidence angle = C:

$i_{wall} = C = 60°$

Inside the rod, ray angle with axis = $r = 90° - C = 30°$

Snell's law for entry from air:

$\sin \theta = \mu \sin r$

$$\sin \theta = \left(\frac{2}{\sqrt{3}}\right) \times \left(\frac{1}{2}\right)$$

$$\sin \theta = \frac{1}{\sqrt{3}}$$

Thus $\theta = \sin^{-1}\frac{1}{\sqrt{3}}$.
FINAL ANSWER: Option (4)

34)

$$\mu = \frac{C}{V} = \frac{3}{1.8} = \frac{5}{3}$$
$$\delta = t\left(1 - \frac{1}{\mu}\right) = 0.25\left(1 - \frac{3}{5}\right) = 0.25 \times \frac{2}{5} = 0.1\ m$$

35) **Given:**

A. Focal length (f) = 30 cm
B. Object distance (u) = -30 cm

Substitute into the mirror formula:
1/(30) = 1/v + 1/(-30)
2/30 = 1/v
v = 15 cm

36)

For real object :

(A)	m = –4	Image is real, inverted and enlarged it is possible in concave mirror only.
(B)	$m = -\frac{1}{4}$	Image is real, inverted and small in size it is possible in concave mirror only.
(C)	m=+3	Image is virtual, errect and enlarged it is possible in concave mirror only.
(D)	$m = +\frac{1}{3}$	Image is virtual, errect and diminished it is possible only in convex mirror.

37)

For any vector in 3D space, its direction cosine [cos α, cos β, cos γ] satisfy :-
$\cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1$
Since $\cos^2\theta_z \geq 0$
it must hold that
$\cos^2\theta_x + \cos^2\theta_y \leqslant 1$
(i) $\Rightarrow \cos^2(60°) + \cos^2(60°)$
$\Rightarrow \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \Rightarrow \frac{1}{2} \Rightarrow (0.5 \leqslant 1)$ possible
(ii) $\cos^2(45°) + \cos^2(60°)$
$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{4} = 0.75$
$\Rightarrow (0.75 \leqslant 1)$ (possible)
(iii) $\cos^2(30°) + \cos^2(45°)$

$\Rightarrow \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow \frac{3}{4} + \frac{1}{2} = 1.25$

1.25 > 1 [Not possible]

(iv) $\cos^2(30°) + \cos^2(30°)$

$= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4} + \frac{3}{4} = 1.5$

(1.5 > 1) (Not possible)

38) **Assertion (A) is false :** Cointial vectors (vector sharing the same starting point) Can point in any direction in 3D space. So they do not all have to lie in the same plane.
Reason (R) is true :- By definition coplanor vectors lie in the same plane.

39) $\vec{A} \cdot \vec{B} = \left(2\hat{i} + \sqrt{7}\hat{j}\right) \cdot \left(5\hat{i} + \sqrt{7}\hat{j} - 3\hat{k}\right) = 17$

$\vec{B} - \vec{A} = 3\hat{i} - 3\hat{k}$

Required vector

$= \frac{\vec{B} - \vec{A}}{\left|\vec{B} - \vec{A}\right|} \cdot \left(\vec{A} \cdot \vec{B}\right) = \frac{3(\hat{i} - \hat{k})}{3\sqrt{2}} 17$

$= \frac{17}{\sqrt{2}}(\hat{i} - \hat{k})$

40)

Given here $\left|\vec{A}\right| = \left|\vec{B}\right| = \left|\vec{C}\right| = 1$

$\Rightarrow \vec{A} + \vec{B} + \vec{C} = \vec{0}$

$\Rightarrow \left(\vec{A} + \vec{B} + \vec{C}\right)^2 = 0$

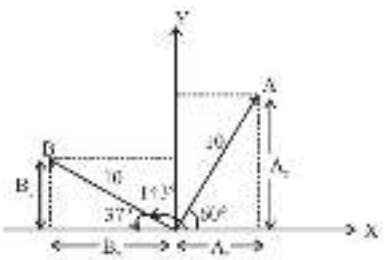
$\Rightarrow \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} + \vec{C} \cdot \vec{C} + 2\left(\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}\right) = 0$

$\Rightarrow A^2 + B^2 + C^2 + 2\left(\vec{A} \cdot \vec{B} \cdot + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}\right) = 0 \quad [\because \vec{A} \cdot \vec{A} = A^2, \vec{B} \cdot \vec{B} = B^2, \vec{C} \cdot \vec{C} = C^2]$

$\Rightarrow 1 + 1 + 1 + 2\left(\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}\right) = 0$

$\Rightarrow \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A} = -\frac{3}{2}$

Final answer: option (2)

41) 

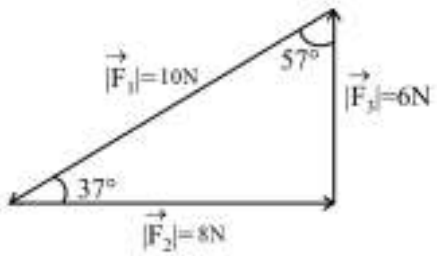
Distance of A from x axis = A_y

$\Rightarrow A_y = 10 \sin 60° = 5\sqrt{3}$

distance of B from x axis = B_y

$\Rightarrow B_y = 10 \sin 37° = 6$
distance of A from y axis = A_x
$\Rightarrow A_x = 10 \cos 60° = 5$
distance of B from y axis = B_x
$\Rightarrow B_x = 10 \cos 37° = 8$

42) $a^2 + b^2 + 2ab\cos\theta = a^2+(-2b)^2 -2(a)(2b)\cos\theta$
$b^2 + 2ab\cos\theta = 4b^2 - 4ab\cos\theta$
$6ab\cos\theta = 3b^2$
$\frac{a}{b} = \frac{1}{2\cos\theta}$
$\left(\frac{a}{b}\right)_{min} = \frac{1}{2(\cos\theta)_{max}} = \frac{1}{2}$
$\left(\frac{a}{b}\right)_{min} = \frac{1}{2}$
possible value of $\left(\frac{a}{b}\right) \geq \frac{1}{2}$



43)
Angle between $\vec{F}_1 \& \vec{F}_2 = 180° - 37° = 143°$

44) $1 = \sqrt{(0.4)^2 + (0.8)^2 + c^2}$
$1 = 0.16 + 0.64 + c^2$
$c^2 + 0.80 = 1$
$c^2 = 1 - 0.80$
$c^2 = 0.20$
$c = \sqrt{0.20}$
$c = \sqrt{\frac{20}{100}} = \frac{1}{\sqrt{5}}$

45) $\vec{a} = \left(2\hat{i} - \hat{j} + \hat{k}\right)$
$\vec{b} = \left(\hat{i} + 2\hat{j} - 3\hat{k}\right)$
$\vec{c} = \left(3\hat{i} + p\hat{j} - 5\hat{k}\right)$
Three vector $\vec{a}, \vec{b}$ and $\vec{c}$ are coplaner if their scalar triple product is equal to ZERO.

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & p & 5 \end{vmatrix} = 0$$

$2 [10 + 3p] +1 [5 + 9] + 1 [p - 6] = 0$

$20 + 6p + 14 + p - 6 = 0$
$7p = -28$
$\boxed{p = -4}$

CHEMISTRY

46) Henry's constant increases with the increase in temperature. Also the value of k_H for every gas is different of different nature of interactions of every gas with the solvent.

According to Henry's law; $p = K_H \times \frac{n_{gas}}{n_{gas} + n_{liquid}}$

⇒ as k_H increases at constant pressure; the mole-fraction of soluble gas decreases.

47) As per Henry's law

$$x_{Ar} = \frac{P_{Ar}}{K_H} = \frac{0.403}{40.3 \times 10^3} = 1 \times 10^{-5}$$

$$\Rightarrow \frac{n_{Ar}}{n_{Ar} + n_{H_2O}} = 1 \times 10^{-5}$$ [1 litre of H_2O= 55.56 mol]

$$\Rightarrow \frac{n_{Ar}}{55.56} = 1 \times 10^{-5}$$

$$\Rightarrow n_{Ar} = 55.56 \times 10^{-5} \text{ mol}$$

48) $P_T = P_A^o X_A + P_B^o X_B = \frac{1}{4} \times P_A^o + \frac{3}{4} \times 3P_A^o$

$$= \frac{10}{4} P_A^o = \frac{5}{2} P_A^o$$

$$P_T . Y_A = P_A^o x_A$$

$$\Rightarrow y_A = \frac{P_A^o \times \frac{1}{4}}{\frac{5}{2} P_A^o} = \frac{1}{10}$$

49) $\frac{P^0 - 0.95P^0}{P^0} = X_B$

$X_B = \frac{1}{20}$

$$\frac{\frac{B}{mm_B}}{\frac{B}{mm_B} + \frac{A}{mm_A}} = \frac{1}{20}$$

$\frac{A}{B} = \frac{507}{1} \left[\frac{mm_B}{mm_A} = \frac{1}{0.3}\right]$

50) **Question Explanation:**
Relation between P_0, P_S, N_1 and N_2
Concept:
This question is based on relative lowering in vapour pressure.

Solution:
R.L.V.P.
$\frac{P_0 - P_S}{P_0} = X_{solute}$
$\frac{P_0 - P_S}{P_0} = N_2$
$P_0 - P_S = N_2 P_0$
Final Answer:
The correct option is (2)

51) $T_b = \Delta T_b + T_b°$ $T_f = T_f° - \Delta T_f$ □ ($T_b° = 100° C$ $T_f° = 0°C$)
$T_b - T_f = 105 \Rightarrow \Delta T_b + 100 - (-\Delta T_f) = 105$
$\Rightarrow \Delta T_b + \Delta T_f = 5$
$\Rightarrow 2.5m = 5 \Rightarrow m = 2$ molal
$m = \frac{W \times 1000}{180 \times 100} \Rightarrow W = 36$ gm

52) Addition of NaCl to water decrease freezing point but produce water of lower temp which prevent melting of ice cream.

53) Methyl alcohol + Water = +ve deviation
Hence BP decreases
A is false, but R is true.

54) osmotic pressure

55) **Explain Question :**
Boiling of non volatile solute depends on
Concept: Elevation in boiling point ΔT_b
Solution:
When a non-volatile solute A is dissolved in a pure solvent B , its vapour pressure decreases and hence the boiling point increases. The difference
ΔT_b of boiling points of the solution and pure solvent is called elevation in boiling point.
If T_b^0 is the boiling point of pure solvent and T_b is the boiling point of the solution then, $T_b > T_b^0$ and the elevation in boiling point $\Delta T_b = T_b - T_b^0$
Experiments have shown that for dilute solutions, the elevation of boiling point (ΔT_b) is directly proportional to the molal concentration of the solute in a solution. Thus
$\Delta T_b \propto m$
or $\Delta T_b = K_b.m$
where K_b = boiling point elevation constant or molal elevation constant or ebulioscopic constant.
Molal elevation constant is characteristic of a particular solvent and can be calculated from the thermodynamical relationship.
$K_b = \frac{RT_b^2 M}{1000 \Delta H_{vap}}$
Where, R is molar gas constant =2cal/mol-K,M= molar mass of solvent

Final Answer = Option (4)
Boiling point of an ideal liquid solution containing non-volatile solute depends on all the given factors. hence all options are correct.

56) K_b of CCl_4 > Kb of CS_2 both non polar.
$\therefore K_b \propto T_{BP} \propto$ Mole mass

57) $2S \rightleftharpoons S_2$
1–α α/2
$\Rightarrow n = 1 - \frac{\alpha}{2}$
$\Delta T_b = k_b \times i \times m \Rightarrow \Delta T = k_b(1 - \frac{\alpha}{2}) \times 1$
$\frac{\Delta T_b}{k_b} = 1 - \frac{\alpha}{2} \Rightarrow \alpha = \frac{2(k_b - \Delta T)}{k_b}$

58)

(P) Acetone + $CHCl_3$ –ve devation from Raoult's law $\Delta S > 0$
$\Delta H < 0$ $\Delta V < 0$
Maximum Boiling Azeotropes.

(Q) Ethanol + Water +ve Deviation from Raoult's law $\Delta S > 0$
$\Delta H > 0$ $\Delta V > 0$ Minimum Boiling Azeotropes

(R) $C_2H_5Br + C_2H_5I$ Ideal solution No Azeotropes
$\Delta H = 0$ $\Delta V = 0$

(S) Acetone + Benzene +ve Deviation from Raoults law
$\Delta H > 0$ $\Delta V > 0$ $\Delta S > 0$
Minimum Boiling Azeotropes

59) $\frac{[Diluted milk]}{[Concentated milk]} = \frac{0.2}{0.5}$
Milk : Water = 2 : 3
Hence 3 cups of H_2O must be added to 2 cups of pure milk.

60) The vapour pressure of mixture (= 600 mm Hg) is greater than the individual vapour pressure of its constituents (Vapour pressure of CS_2 = 512 mm Hg, acetone = 344 mm Hg). Hence, the solution formed shows positive deviation from Raoult's law.
$\Rightarrow$ (1) $\Delta_{Sol}H > 0$,
(2) Raoult's law is not obeyed
(3) $\Delta_{sol.}$ Volume > 0
(4) CS_2 and Acetone are less attracted to each ether than to themselves.

61) **For A :** 100 gm solution → 2 gm solute A

$\therefore$ Molality = $\frac{2/M_A}{0.098}$

For B : 100 gm solution → 8 gm solute B

$\therefore$ Molality = $\frac{8/M_B}{0.092}$

$\because (\Delta T_B)_A = (\Delta T_B)_B$

$\therefore$ Molality of A = Molality of B

$\therefore \frac{2}{0.098 M_A} = \frac{8}{0.092 M_B}$

$\frac{2}{98} \times \frac{92}{8} = \frac{M_A}{M_B}$

$\frac{1}{4.261} = \frac{M_A}{M_B}$

$\therefore M_B = 4.261 \times M_A$

62)

$P_s = 120 - 80\ X_B$

For pure B, $X_B = 1$, $Ps = p°_B = 40$

For pure A, $X_B = 0$, $Ps = p°_A = 120$

63) **Solution:**

Molality is defined as no. of moles of solute dissolved in 1000 g of solvent

$m = \frac{n_B}{M_w \times w_A(g)} \times 1000$

$= \frac{18 \times 1000}{180 \times 1000} = 0.1\,m$

Final Answer: Option (4)

64) $X_A = \frac{n_A}{n_A + n_B}$

$n_{glycerine} = \frac{w}{M_w} = \frac{46}{92} = 0.5 = n_A$

$n_{H_2O} = \frac{w}{M_w} = \frac{36}{18} = 2 = n_B$

$X_A = \frac{n_A}{n_A + n_B} = \frac{0.5}{2.5} = 0.2$

65) **Solution:**

$M = \frac{w}{M_w \times V(L)} = \frac{5}{34 \times 0.1} = 1.5$

Final Answer: Option (2)

66)

Higher is (i × c); higher will be ΔT_f, So, lower will be T_f

i × c for $KNO_3 = 2 \times 0.05 = 0.1$

$BaCl_2 = 3 \times 0.04 = 0.12$
Sucrose $= 1 \times 0.140 = 0.14$
$CuSO_4 = 2 \times 0.075 = 0.15$

67)

For S_1; $i_1 \times C_1 = 2 \times 0.1 = 0.2 \rightarrow$ Hypertonic
For S_2; $i_2 \times C_2 = 2 \times 0.05 = 0.15 \rightarrow$ Hypotonic

68)

$P_S = P_A^0 \cdot x_A + P_B^0 \cdot x_B$
$SP = P_A^0 \times 0.2 + 0.8 \times P_B^0$
SP = SR + SQ

69)

$(\Delta Tb)_I = K_b \times \frac{0.1}{m_P \times 100} \times 1000$ .....(i)
$(\Delta Tb)_{II} = K_b \times \frac{0.1}{m_Q \times 100} \times 1000$ .....(ii)
I/II
$\frac{(\Delta Tb)_I}{(\Delta Tb)_{II}} = \frac{M_Q}{M_P}$
$\frac{0.2}{0.4} = \frac{M_Q}{M_P}$ $\boxed{\frac{M_P}{M_Q} = \frac{2}{1}}$

70) $\pi = \frac{n}{V} \times RT \quad \Rightarrow \quad \pi = \frac{W}{M} \times \frac{RT}{V} = \frac{CRT}{M}$
(C ⇒ in g/L)
$\therefore \log p = \log C + \log\left(\frac{RT}{M}\right)$
Thus OA $= \log(x \times 10^{-2}) = \log\left(\frac{RT}{M}\right)$
$\Rightarrow x \times 10^{-2} = \frac{RT}{M}$
$\Rightarrow x \times 10^{-2} = 0.0821 \times \frac{300}{300}$
$\Rightarrow x \times 10^{-2} = 8.21 \times 10^{-2}$
$\Rightarrow x = 8.21$

71) For monobasic acid HA

HA	⇌	H^+	+	A^-
1		0		0
(1–α)		α		α

From Ostwald dilution law,
Now, $\Delta T_{nor} = K'_f \times \text{molality} = 1.86 \times 0.025$
$= 0.0465$

$$i = \frac{\text{Observed } \Delta T}{\text{Calculated } \Delta T} = 1 + \alpha$$

$$i = \frac{0.06}{0.0465} = 1 + \alpha$$

$\therefore \alpha = 0.29$ and $C = 0.025$ M
By Eq. (1),

$$K_\alpha = \frac{0.025 \times (0.29)^2}{(1 - 0.29)} = 2.96 \times 10^{-3}$$

72)

Explain Question : We need the volume of water required to dilute 10 mL of 10 N HCl to 0.1 N.
Concept : This question is based on Dilution formula: $N_1V_1 = N_2V_2$
Solution :
$N_1 = 10$ N, $V_1 = 10$ mL, $N_2 = 0.1$ N
$V_2 = (N_1V_1) / N_2 = (10 \times 10) / 0.1 = 1000$ mL
Total volume needed = 1000 mL
Water added = 1000 – 10 = 990 mL
Final answer : 1

73) $$\frac{P_A^0 - P_A}{P_A^0} = X_B$$

74) Those concentration unit which do not involve volume are temperature independent molality(m), mole fraction(x) etc.

75)

Question Explanation : To identify the solution shows +ve deviation
Concept: Non ideal solution
Solution:
(i) $CHCl_3$ + Acetone ⇒ –ve deviation
(ii) CS_2 + Acetone ⇒ +ve deviation
(iii) Ethanol + Acetone ⇒ +ve deviation
(iv) phenol + Anilene ⇒ –ve deviation
Final answer: Option (2)

76) **Explain Question :** Find mass of glucose per litre that gives same osmotic pressure as 4g urea per litre (same T → isotonic).
Concept : This question is based on Osmotic pressure ($\pi = iCRT$, $i = 1$), isotonic ⇒ equal molarity ($\pi_1 = \pi_2$).
Solution :
$\pi_1 = \pi_2$

$C_1 = C_2$
Molar mass: urea $CO(NH_2)_2$ = 60 g·mol^{-1}; glucose $C_6H_{12}O_6$ = 180 g·mol^{-1}.
Urea molarity = 4/60 = 1/15 ≈ 0.0667 mol·L^{-1}.
Same molarity for glucose ⇒ $M_{(glucose)}$ = 0.0667 M
Or concentration (in g/L) = 0.0667 × 180 = 12 g / L.
Final answer : 3

77)

Explain Question :- The question asks to calculate the percentage dissociation of an electrolyte, given its normal and observed molar masses.
Concept :-This question is based on The van't Hoff factor (i) : i = 1 + (n –1)α.
Solution :-
$$i = \frac{\text{Normal Molar Mass}}{\text{Observed Molar Mass}} = \frac{250}{100} = 2.5$$
$XY_2 \rightarrow X^{2+} + 2Y^-$
So, n = 1 + 2 = 3
Using the formula i = 1 + (n–1)α
2.5 = 1 + (3 – 1)α
2.5 = 1 1 + 2α
1.5 = 2α
$$a = \frac{1.5}{2} = 0.75$$
Final answer :- (1)

78) For non-ideal solutions
(1) $P_S > x_A P_A^0 + x_B P_B^0$
$P_S < x_A P_A^0 + x_B P_B^0$
(2) ΔH = –ve (Negative deviation)
ΔH = +ve (For positive deviation)
(3) ΔV > 0 (For positive deviation)

79) **Explain Question :** We need to find how many moles of solute particles (van't Hoff factor i) are present per mole of $CoCl_3 \cdot 6NH_3$, using freezing point depression data.
Concept : This question is based on Freezing point depression: $\Delta T_f = i \times K_f \times m$
Solution : Moles of solute = 1.1 g / 267 g mol^{-1} = 0.00412 mol
Mass of solvent = 100 g = 0.100 kg
Molality (m) = 0.00412 / 0.10 = 0.0412 mol kg^{-1}
Observed ΔT_f = 0.29 °C
Now, $i = \Delta T_f / (K_f \times m)$
= 0.29 / (1.86 × 0.0412) = 3.78 ≈ 4
Final answer : 4

80)

$CCl_4 + C_6H_6$ = +ve deviation
So $V_{mix} > V_1 + V_2$

$V_{mix} > 30 + 10$
$V_{mix} > 40$ ml

81)

Explain Question – Which of the following is correct regarding depression in freezing point.

Concept – This question based on colligative property.

Solution –

(a) Vapour pressure of solution is less than that of pure solvent: This is correct. The presence of solute particles reduces the number of solvent molecules at the surface, thus lowering the vapor pressure of the solution compared to the pure solvent.
(d) Only solvent molecules solidify at the freezing point: This is also correct. When a solution freezes, typically only the solvent crystallizes out, leaving the solute in the remaining liquid phase.

Final Answer – Option (1)

82) **Explain the Question :** The question evaluates the causal relationship between adding a non-volatile solute to a solvent and the resulting decreases in its freezing point temperature.
Concept : This question is based on depression in freezing point.
Solution : Both assertion and reason is true and the reason is the correct explanation of the assertion.
 After adding a non-volatile solute, the freezing point of a solvent is depressed because the presence of solute particles lowers the vapour pressure of the liquid solvent so the solid-liquid equilibrium is reached only at a lower temperature.
Final Answer : (1)

83)

Question Explanation:
Characteristics of non-ideal solutions and Raoult's law.

Concept:
Negative Deviation from Raoult's Law.

Solution:
A solution becomes hot when the mixing process is exothermic ($\Delta H_{mix} < 0$). This happens when the new interactions between different molecules (A-B) are stronger than the original ones (A-A and B-B), leading to a negative deviation. However, stronger interactions pull molecules closer together, causing the volume to decrease ($\Delta V_{mix} < 0$), not increase.
Calculation For Negative Deviation: $\Delta H_{mix} < 0$ (Exothermic/Hot)
$\Delta V_{mix} < 0$
• Volume change: (Decrease in volume)
• The Reason states volume increases, which is false.

Final Answer: Option (3)

84)

Question Explanation: The question checks the correctness of the Assertion and Reason related to expressing pollutant concentration in ppm.

Concept: Concentration terms; parts per million (ppm) is used for very dilute solutions.

Solution:

Assertion: Concentration of pollutants in water or air is often expressed in ppm.
→ True, because pollutants are usually present in very small amounts.

Reason: ppm can be expressed as mass/mass, volume/volume, or mass/volume.
→ True, these are valid ways to express ppm.

$$\text{ppm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$

$$\text{ppm} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 10^6$$

$$\text{ppm} = \frac{\text{mass of solute}}{\text{volume of solution}} \times 10^6$$

However, the Reason only defines forms of ppm; it does not explain *why* pollutants are expressed in ppm.
Both Assertion & Reason are true, but Reason is not the correct explanation of the Assertion.

Final Answer: Option (2)

85)

Question Explanation: Match each type of solution (solute–solvent) with its correct real-life example.

Concept: Classification of solutions based on physical state of solute and solvent.

Solution:

(A) **Liquid in gas** → Chloroform (liquid) mixed with nitrogen gas → **(iii)**

(B) **Solid in solid** → Copper (solid) dissolved in gold (solid) → **(i)**

(C) **Liquid in solid** → Amalgam of mercury (liquid) with sodium (solid) → **(iv)**

(D) **Solid in liquid** → Glucose (solid) dissolved in water (liquid) → **(ii)**

Correct match - (A)→(iii), (B)→(i), (C)→(iv), (D)→(ii)

Final Answer: Option (2)

86)

Explain Question:
We must match each law (Dalton’s law for liquid, and Raoult’s law for solid –liquid) with its

correct expression.

Concept:
Gas mixtures → Dalton; liquid solutions → Raoult; solid-liquid → solute lowering vapour pressure.

Solution:
Dalton's law → partial pressure = mole fraction × total pressure → matches (Q).
Raoult's law (liquid-liquid) → $P_A = P_A^0 X_A^0$ → matches (P).
Raoult's law (solid-liquid) → relative lowering of vapour pressure formula matches (R).
Correct matching: (i)-(Q), (ii)-(P), (iii)-(R)

Final answer: Option (3)

87)

Explain Question: Match list-I (solutes) with list-II (ratio of depression in freezing point) and select the correct answer from the codes given below:
Concept: Depression in freezing point.
Solution:
Van't Hoff values:
Urea, Sugar, Glucose → i = 1
NaCl, KCl → i = 2
Na_2SO_4, K_2SO_4 → i = 3
(A) Urea (1), NaCl (2), K_2SO_4 (3) → ratio 1:2:3 → (b)
(B) KCl (2), NaCl (2), K_2SO_4 (3) → ratio 2:2:3 → (c)
(C) Urea (1), Sugar (1), Glucose (1) → ratio 1:1:1 → (d)
(D) Urea (1), KCl (2), Glucose (1) → ratio 1:2:1 → (a)
Correct code: A-b, B-c, C-d, D-a.
Final answer: Option (1)

88) **Explain question:**
To select correct option for the Osmosis.

Concept:
Osmosis

Solution:
Osmosis is spontaneous movement of solvent molecules across SPM from pure solvent to solution side or form low conc. solution to higher conc. solution side.

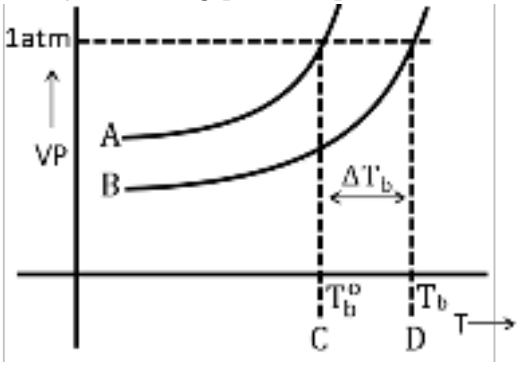
Final answer: Option (2)

89)

Explain Question: To select correct option for vapour pressure vs temperature curve.

Concept: Elevation in boiling point.

Solution:
Vapour pressure of the solution having non volatile solute is less than of the solvent, so the boiling point of solution is greater than of solvent.
A → Vapour pressure of pure solvent.
B → Vapour pressure of solution having non volatile solute.
D → T_b ⇒ Boiling point of solution.
C → T_b^0 ⇒ Boiling point of pure solvent.



Final Answer: Option(2)

90) **Explanation:** The question asks you to determine the type of deviation from Raoult's law (or ideality) that a solution of acetone in ethanol exhibits.

Given data: Solution: Acetone in ethanol

Concept: Acetone and ethanol show positive deviation from Raoult's law because the A-A and B-B interactions (hydrogen bonding) are stronger than the A-B interactions.

Final answer: positive deviation from Raoult's law.

Correct option: 1

BIOLOGY

91) NCERT Pg.117

92) NCERT Pg. # 116-118

93) The correct option is: **(2) A - a,c,d ; B - b,e,f**

Explanation:

A. **A - (a) Acid-soluble pool:** Contains small molecules, such as **amino acids, sugars, nucleotides (d)**, with molecular weights ranging from **18 to 800 daltons (c)**.
B. **B - (b) Acid-insoluble pool:** Contains larger molecules like **proteins, nucleic acids, polysaccharides (f)**, with molecular weights **greater than 1000 daltons (e)**.

Hence, option (2) is correct.

94) NCERT Pg.#110

95) **Question Explanation:** Match proteins with their correct characteristics/functions.
Concept: Biomolecules (Proteins)
Solution:
(A) RuBisCO → Most abundant protein in biosphere → (III)
(B) Insulin → Hormone → (IV)
(C) Collagen → Most abundant protein in animal world → (I)
(D) Trypsin → Enzyme → (II)
Final Answer: Option (2)

96)

Explain Question:
The question asks us to correctly match the levels of protein structure (List-I) with their examples/features (List-II).

Concept:
Protein Structure.

Solution:

A. A. Primary structure → III. Polypeptide chain
B. B. Secondary structure → IV. Alpha helix and β sheet
C. C. Tertiary structure → II. Disulphide bonds)
D. D. Quaternary structure → I. Human haemoglobin

Final Answer: Option (1)

97) NCERT Pg. # 108

98)

Question Explanation: The question asks about the monomer unit and glycosidic linkage present in cellulose.

Concept: Cellulose Structure

Solution:
Cellulose is a linear homopolysaccharide made up of β-D-glucose units. These glucose units are joined by β-1→4 glycosidic linkages.

Final answer: Option (1) (A) β-D-glucose; (B) β-1′–4″

99) NCERT XI Pg # 106

100)

Module No. 3, Pg. # 5, 6

101) NCERT Page NO. 133-150

102) NCERT Pg.# 142

103) NCERT, Pg # 139,140

104)

NCERT XI Pg # 143-144

105) NCERT Pg. # 146, 147

106) NCERT Pg. # 145

107) NCERT Pg. # 145

108) NCERT Pg. # 134, 135

109) NCERT, Pg. # 138

110) NCERT, Pg. # 136, 137

111) NCERT, Pg. # 136 & 138, fig. 11.2, 13.2(a)

112)

NCERT-XI, Pg. # 146

113) NCERT Page No. 139

114) **Question Explanation:** Incorrect statement on C_3 and C_4 plants.
Concept: C_3 and C_4 Photosynthesis
Solution: C_3 and C_4 plants have different mechanisms for carbon fixation. Due to the efficient CO_2 concentrating mechanism of C_4 plants, they show a different response to CO_2 levels compared to C_3 plants. C_4 plants saturate at lower CO_2 concentrations, while C_3 plants require a much higher CO_2 concentration to reach saturation. Therefore, they do not respond similarly to CO_2 concentration.
Final Answer: The C_3 and C_4 plants respond similarly to CO_2 concentration.

115)

NCERT-XI, Pg. # 149, 150

116) NCERT Pg. # 144

117)

NCERT-XI, Pg. # 164, 165

118)

Target Module No.-3, Pg. # 126, 129

119) NCERT XI Pg.# 179 (E & H)

120)

NCERT (XI) Pg # 164

121) NCERT (XIth) Pg. # 160

122) NCERT XI Page # 156-159

123)

NCERT-XI, Pg. # 156, 157, 159

124)

Target Module No.-3, Pg. # 136

125) NCERT (XIth) Pg. # 163

126)

Target Module No.-3, Pg. # 133

127) NCERT Pg.# 160

128) NCERT Pg. # 158

129) NCERT XI Pg. # 156

130) **Question Explanation:** Verifying statements on cellular respiration.
Concept: Cellular Respiration
Solution: Statement A is correct: Pyruvic acid is converted to acetyl CoA before it enters the TCA (Krebs) cycle. Statement B is incorrect: Succinyl CoA is a four-carbon intermediate, not five. Statement C is correct: The TCA cycle occurs in the mitochondrial matrix. Statement D is correct: Glycolysis takes place in the cytosol.
Final Answer:(4) A, C, D

131)

NCERT-XI, Pg. # 162

132) NCERT Pg. No. # 158

133) NCERT Pg. No. # 139, 177, 176, 158

134) NCERT Pg. # 168, 169

135) NCERT, Pg # 176,177

136) NCERT XI Pg. # 174,175

137)

NCERT-XI, Pg. # 177

138) **Explain Question :** Identify the correct statements about functions of different plant hormones.

Concept : This question is based on PGRs

Solution:
(a) Correct — cytokinins delay senescence.
(b) Incorrect — apical dominance is due to auxin.
(c) Correct — ethylene can enhance seed germination by breaking dormancy.
(d) Incorrect — immature leaf fall caused by ethylene/ABA, not gibberellins.

Final Answer :
Option(1)

139)

NCERT Pg. # 171

140) NCERT XI Pg # 167

141)

Solution correction :

Explain Question: The question asks to choose the correct options

Concept: Conditions for growth

Solution:

(I) O_2 helps release metabolic energy via respiration.
(II) Nutrients, while mainly for synthesis, can indirectly contribute to energy processes.
(III) Optimum temperature is essential for proper growth.
(IV) Light and gravity act as environmental signals affecting growth.

So, the correct answer is: All are correct

Final answer: Option (1).

142)

NCERT-XI, Pg. # 172

143) **Explain Question :**
The question is about the sigmoid or S-shaped growth curve, which is a typical pattern of growth observed in many organisms.

Concept :
This question is based on Growth Rates - exponential growth.

Explanation:
"The exponential growth cannot be sustained indefinitely because the nutrient supply becomes limited eventually."

Final answer:3

144) NCERT XI Page # 175

145) NCERT Pg. # 160

146) NCERT Pg # 117-118

147)

NCERT-XI, Pg. # 117

148) NCERT Pg. # 115, 116

149)

Module No.-3, Pg. # 89

150)

NCERT Pg.# 158

151) **Solution :** The stroma of the chloroplast contains enzymes required for the synthesis of both carbohydrates and proteins. It also contains small, double-stranded circular DNA molecules and ribosomes. Oxidative phosphorylation takes place in mitochondria, not chloroplasts.
Final Answer : (3)

152) NCERT XI P.NO. 91

153) NCERT-XI, Pg. # 99

154) NCERT-XI, Pg. # 95, 96

155) NCERT-XI, Pg. # 100

156) NCERT XI Pg. # 93,94

157)

NCERT-XI, Pg. # 95

158) NCERT Class-XI : Page No. 100, 101, 123, 124

159) NCERT-XI, Pg # 124(E)

160) NCERT XI P. NO . 121-122

161) NCERT XI Pg. No. # 167 (Topic - Meiosis - Ist point)

162) NCERT-XI, Pg. # 165,169, 170

163)

NCERT XI Pg. No. # 121

164)

NCERT-XI, Pg. # 126
The correct answer is:
Option 3: Both statements are correct.
Explanation:
1. Statement I: "Synaptonemal complex is a structure involved in the pairing of homologous chromosomes." **(Correct)**
• The synaptonemal complex is a protein structure that forms between homologous chromosomes during zygotene (prophase I of meiosis I).
• It helps in synapsis, ensuring proper chromosome alignment and recombination.
2. Statement II: "Crossing over occurs between non-sister chromatids of homologous chromosomes." **(Correct)**
• Crossing over happens during pachytene (prophase I of meiosis I).
• It involves the exchange of genetic material between non-sister chromatids of homologous chromosomes.
• This genetic recombination increases genetic variation in gametes.
Final Answer:
Option 3: Both statements are correct.

165)

NCERT XI, Pg. # 122

166)

Explain Question – The question evaluates two statements related to cymose inflorescence and solitary flowers and asks to choose the correct option.

Concept –Morphology of Flowering Plants-Inflorescence

Solution –

A. In cymose inflorescence, the main axis terminates in a flower, therefore growth is limited, and flowers are arranged in basipetal order. Statement-I is correct.

B. When the shoot tip transforms into a flower, further growth stops and the flower is solitary

Statement-II is also correct.

Final answer – Option 4

167)

NCERT, Pg. # 69, 70 (5.3)

168) NCERT, Pg. # 75 (5.5.1.4)

169) NCERT, Pg. # 76 (5.6)

170) NCERT, Pg. # 72

171) NCERT (XI) Pg # 61

172)

NCERT XI, Pg # 67

173)

NCERT XI Pg. # 65

174)

NCERT XI, Pg # 73

175)

NCERT XI, Pg # 91

176) NCERT -XI, Pg. # 87

177) NCERT-XI, Pg. # 91, 92

178)

Question Explanation: Identifying characteristics of bulliform cells.
Concept: Monocot Leaf Anatomy
Solution: Bulliform cells are a type of large, thin-walled, and colorless epidermal cells found on the adaxial (upper) surface of monocot leaves like grasses. When they are turgid, the leaf is unrolled. When they become flaccid due to water loss, they cause the leaf to roll inwards, which reduces the surface area exposed to the sun and minimizes transpiration.
Final Answer: They are large, empty, colourless cell

179)

NCERT XI Page No. 92

180) NCERT-XI Pg. # 75